
FLAG: Flow Policy MaxEnt-RL by Latent Augmented Guidance

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Abstract

1 Maximum entropy reinforcement learning (MaxEnt-RL) enables robust exploration,
2 yet practical implementations often restrict policies to simple Gaussians. While re-
3 cent approaches incorporate expressive generative policies via importance-weighted
4 supervised learning, they are prone to importance weight collapse, which limits
5 their scalability in high-dimensional action spaces. Our key insight is to mitigate
6 this limitation by localizing the sampling region, avoiding the weight degeneracy
7 induced by importance sampling over the entire action space. To instantiate
8 this insight, we introduce **FLAG** (Flow policy with Latent-Augmented Guidance).
9 FLAG augments the state space with a flow latent variable and optimizes a provably
10 consistent proxy MaxEnt-RL objective. We empirically demonstrate that FLAG
11 enables expressive policy optimization with limited importance samples and scales
12 to high-dimensional control tasks. Furthermore, FLAG achieves state-of-the-art
13 performance across challenging benchmarks.

14 1 Introduction

15 Maximum entropy reinforcement learning (MaxEnt-RL) enhances conventional reward maximization
16 with an entropy regularization term, enabling robust decision-making and persistent exploration [50,
17 45, 19, 20]. By encouraging stochasticity in the policy, MaxEnt-RL can represent multiple near-
18 optimal behaviors through expressive action distributions [19]. Most existing methods parameterize
19 the policy as a Gaussian distribution due to its optimization efficiency [20, 7, 32]. However, the
20 unimodal nature of Gaussian policies fundamentally limits their ability to capture the complex optimal
21 policies induced by the MaxEnt-RL objective. To address this expressivity bottleneck, recent studies
22 have introduced diffusion- and flow-based generative policies, achieving strong performance on
23 high-dimensional continuous control benchmarks [47, 11].

24 Despite their empirical success, these approaches typically require Backpropagation Through Time
25 (BPTT) across multiple generative steps, which can suffer from numerical instability [6, 34]. To avoid
26 these issues, a separate line of work [16, 27, 12] adopts an Expectation-Maximization (EM, [29])
27 framework in the vein of MPO [2], casting policy optimization as supervised learning on the target
28 distribution. While this strategy circumvents BPTT, it still relies on importance sampling (IS, [41])
29 because sampling from the MaxEnt target distribution is intractable. In high-dimensional spaces,
30 the proposal–target mismatch can cause importance weights to explode, exacerbating the vanishing
31 support problem [30, 42]. Prior works [16, 27, 12, 13] heuristically constrain IS weights for stability,
32 but these post-hoc corrections do not directly increase the probability of sampling target-relevant
33 actions. As a result, IS over the entire action space, which we refer to as *global IS*, remains prone to
34 weight degeneracy, leading to poor sample efficiency and sparse supervision insufficient for tracking
35 non-stationary targets in online RL. We illustrate this sampling inefficiency with a didactic multigoal
36 example under limited sample budgets N (Figure 1), where global IS baselines fail to capture the

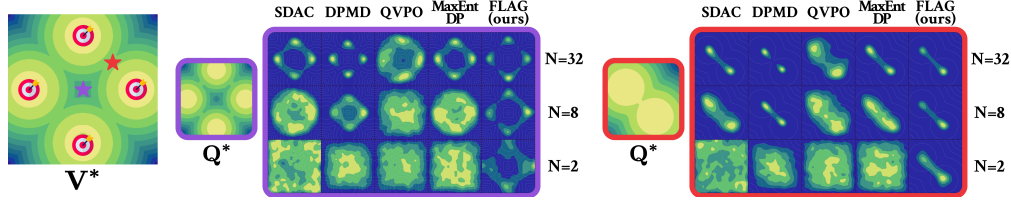


Figure 1: **Comparisons with global policy sampling methods in multi-goal environment.** The multi-goal environment [19] tasks a point mass with navigating to one of four symmetrically placed targets $\star$. We plot the optimal value function V^* and Q-functions Q^* at two different states ($\star$ & $\star$), along with policies at each state. While other methods fail to learn with fewer samples ($N \leq 8$), **FLAG** captures optimal and multi-modal behaviors leveraging latent-augmented guidance, even in limited importance sample budget commonly arising in high-dimensional action spaces. Additional details on the environment and setup are presented in Appendix G.1.

target mode as N decreases from 32 to 2. We observe the same trend in high-dimensional control (Figure 3).

Our key insight is to mitigate the sparsity of global IS by jointly localizing the proposal and target distributions. By conditioning on a latent variable, both distributions are restricted to the same local region, allowing IS to operate locally rather than over the full action space, which we refer to *local IS*. To achieve this, we leverage the deterministic nature of flow-matching models, which induces a fixed mapping between latent vectors and actions. Based on this property, we construct a Gaussian *local policy* centered on the flow’s output, which admits a formal definition of a latent-augmented MDP where the standard state space is extended to include the latent variable. Within this framework, the *global policy* is recovered by marginalizing over the latent space, providing a theoretical foundation for optimization via the local IS rather than the global IS.

To instantiate this insight, we propose **FLAG** (Flow policy MaxEnt-RL by Latent Augmented Guidance), which trains the flow policy in a supervised manner on the latent-augmented MDP. Specifically, we introduce a cross-entropy-based proxy MaxEnt objective that avoids the intractable entropy computation of the global policy, and show that it remains consistent with the original MaxEnt objective when the local policy variance is properly controlled during training. We then optimize the proxy objective with an EM-style procedure, using the resulting locally updated policies as latent-conditioned supervised targets for the flow policy, which we call *latent-augmented guidance*. This latent-conditioned local IS allows FLAG to capture the target mode with only $N = 2$ samples in Figure 1, suggesting better scalability to high-dimensional action spaces (Figure 3).

In summary, our contributions are threefold:

- We introduce the latent-augmented MDP to derive local IS and derive a proxy MaxEnt objective on this augmented MDP, with a theoretical consistency with respect to the original objective.
- We propose FLAG, which optimizes the proxy MaxEnt objective via an EM algorithm where samples are drawn by local IS, and prove that FLAG monotonically improves the proxy objective.
- We empirically demonstrate that FLAG outperforms global IS baselines and achieves state-of-the-art performance across challenging benchmarks.

2 Related Works

Action Gradient. DIPO [49] and QSM [38] leverage Q-gradients ($\nabla_a Q$) to construct supervised policy updates, either by refining replay-buffer actions or matching policy scores. However, their reliance on Gaussian noise exploration motivates a more principled MaxEnt-RL framework.

BPTT-based Actor-Critic. Recent methods treat generative models as parameterized policies within actor-critic frameworks. For example, DACER [47, 48] optimizes Q-values along reverse diffusion, while DIME [11] optimizes Q-values along with a tractable marginal-entropy lower bound. However, these objectives require BPTT through generation, making optimization memory-intensive and numerically unstable. FlowRL [26] avoids BPTT using a single-step policy with implicit optimality guidance, but its objective lacks exploration and can be suboptimal. In contrast, our method derives latent-augmented guidance and optimizes the proxy MaxEnt-RL objective with supervised updates, bypassing BPTT without sacrificing the expressivity of the underlying generative model.

Target Policy Matching. Supervised policy updates in RL have been widely studied under the probabilistic inference framework [24], including EM-based methods such as RWR [37], AWR [36], and MPO [2]. Recent generative-policy extensions, such as RSM [27] and MaxEntDP [16], use reweighted score matching or Q-weighted noise estimation to project policies toward target distributions. However, they rely on global IS, which disperses samples over the full action space and provides sparse supervision in regions important for policy improvement. FLAG instead uses local IS to concentrate samples around each action, yielding denser and more effective policy supervision.

Policy Gradient. FPO [28] and GenPO [14] adapt PPO [40] to generative policies, but are on-policy methods. Our closest off-policy baseline is QVPO [13], which optimizes a weighted diffusion loss to improve the policy. QVPO’s weighted score-matching update relies on a global proposal distribution, inheriting the sparsity issue of global IS, where target-relevant actions are rarely sampled; in contrast, FLAG uses a localized proposal for policy improvement.

3 Preliminaries

3.1 Reinforcement Learning as Probabilistic Inference

Notation. We consider a Markov Decision Process (MDP) defined by the tuple $\langle \mathcal{S}, \mathcal{A}, p, r, \gamma \rangle$, where $\mathcal{S}$ and $\mathcal{A}$ denote the continuous state and action spaces, $p : \mathcal{S} \times \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^+$ represents the transition probability density, $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the reward function, and $\gamma \in [0, 1)$ is the discount factor. We define a stochastic policy $\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^+$, mapping states to probability densities over actions. In the infinite-horizon setting, π induces $p_\pi(\tau)$ and discounted marginals $\rho_\pi(s)$ and $\rho_\pi(s, a)$.

Control-as-inference and MaxEnt-RL. RL admits a probabilistic inference view via binary optimality variables O_t , where $O_t = 1$ indicates (s_t, a_t) is optimal [24]. With $\mathcal{O} = \{O_t = 1\}_{t=0}^\infty$ and $p(\mathcal{O} | \tau) \propto \exp(\sum_t \gamma^t r(s_t, a_t) / \alpha)$ for temperature $\alpha > 0$, Bayes’ rule gives the optimal posterior

$$p^*(\tau | \mathcal{O}) \propto p(s_0) \prod_{t=0}^\infty p(a_t | s_t) p(s_{t+1} | s_t, a_t) \exp\left(\sum_{t=0}^\infty \gamma^t (r(s_t, a_t) / \alpha)\right). \quad (1)$$

Matching $p_\pi(\tau)$ to this posterior via KL-divergence recovers the standard MaxEnt-RL objective [20]:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\tau \sim p_\pi} \left[\sum_{t=0}^\infty \gamma^t (r(s_t, a_t) - \alpha \log \pi(a_t | s_t)) \right]. \quad (2)$$

EM-style Policy Optimization. Introducing a variational policy $q(a | s)$ that shares the initial-state distribution and dynamics with p_π , the trajectory-level KL reduces to action-only terms, yielding the variational lower bound [1, 2]

$$\mathcal{J}(q, \pi) = \mathbb{E}_{\tau \sim p_q} \left[\sum_{t=0}^\infty \gamma^t (r(s_t, a_t) / \alpha) \right] - D_{\text{KL}}(p_q(\tau) \| p_\pi(\tau)). \quad (3)$$

For the current parameterized policy $\pi_k = \pi_{\theta_k}$, the KL-constrained E-step

$$q_k(\cdot | s) = \arg \max_{q \in \Delta(\mathcal{A})} \mathbb{E}_{a \sim q} [Q^{\pi_k}(s, a)] \quad \text{s.t.} \quad D_{\text{KL}}(q(\cdot | s) \| \pi_k(\cdot | s)) \leq \epsilon \quad (4)$$

admits a closed form $q_k(a | s) \propto \pi_k(a | s) \exp(Q^{\pi_k}(s, a) / \lambda_k)$, with $\lambda_k > 0$ the dual variable. The M-step projects q_k back into the parametric class, yielding a weighted maximum-likelihood:

$$\theta_{k+1} \in \arg \max_{\theta} \mathbb{E}_{s \sim \rho_{\pi_k}} \mathbb{E}_{a \sim q_k(\cdot | s)} [\log \pi_\theta(a | s)], \quad (5)$$

typically estimated via importance weighting on samples from π_k .

3.2 Conditional Flow Matching

Flow matching [25] transports a prior p_0 to a target p_1 . It learns a time-conditioned vector field $v : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$. The flow ψ is governed by the following ordinary differential equation (ODE):

$$\frac{d}{dt} \psi^t(x) = v^t(\psi^t(x)) \quad \text{i.e.} \quad \psi^t(x) = \psi^0(x) + \int_0^t v^\tau(\psi^\tau(x)) d\tau. \quad (6)$$

By parameterizing the vector field v with a parameter θ and under the Optimal Transport linear path, the vector field can be directly optimized via the Conditional Flow Matching (CFM) objective:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, x_0, x_1} \left[\left\| v_\theta^t((1-t)x_0 + tx_1) - (x_1 - x_0) \right\|^2 \right], \quad (7)$$

where $t \sim \mathcal{U}[0, 1]$, $x_0 \sim p_0$, and $x_1 \sim p_1$.

4 Method

FLAG targets the optimization of a flow policy guided by local supervision from a Gaussian local policy. First, we introduce a policy parameterization (4.1) that explicitly couples the flow-based anchor actions with a local Gaussian distribution. Next, we lift the original environment to a latent-augmented MDP (4.2). This augmented construction makes the local conditioning variable explicit and establishes that optimizing the local policy in the lifted MDP is mathematically equivalent to optimizing the global MaxEnt policy. On top of this equivalence, we derive an EM-based policy improvement procedure (4.3) that admits the latent-augmented guidance. Finally, we introduce a practical algorithm (4.4) that enables updates of the flow policy via the latent-augmented guidance. All derivations in this section are in Appendix B.

4.1 Policy Parameterization

Our policy parameterization centers the Gaussian on the flow’s anchor $T_\theta(s, z)$, which localizes the effective sampling region around each latent z , providing dense supervision for EM updates and circumvents the importance-weight degeneracy of global proposal sampling. We first define a flow transformation $T_\theta(s, z)$ parameterized by θ that maps a latent vector $z = a^0 \sim p_z$ to an anchor action $T_\theta(s, z) = a^1 = z + \int_0^1 v_\theta(a^\tau, \tau, s) d\tau$ following Eq. (6), inducing the base flow policy $\tilde{\pi}_\theta(a | s) = p_z(z) |\det(\partial T_\theta(s, z)/\partial z)|^{-1}$.

Next, we augment each flow-generated anchor action $T_\theta(s, z)$ with an auxiliary local Gaussian:

$$\hat{\pi}(a | s, z; \theta) = \mathcal{N}(a; T_\theta(s, z), \Sigma). \quad (8)$$

For theoretical development in this paper, we treat Σ as non-learnable, isotropic covariance whose schedule is specified in Section 4.4; a learnable variant is discussed in Appendix D.4. Marginalizing over the prior recovers the global policy:

$$\pi(a | s; \theta) = \int p_z(z) \hat{\pi}(a | s, z; \theta) dz. \quad (9)$$

4.2 Latent augmented MDP and Proxy MaxEnt-RL Objective

In this section, we show that local policy optimization on the augmented MDP is mathematically equivalent to maximizing the original MaxEnt-RL objective in the original MDP. We start by defining the latent-augmented MDP.

Definition 4.1 (Latent-augmented MDP). The latent augmented MDP $\hat{\mathcal{M}}$ is defined by the state $\hat{s} = (s, z) \in \mathcal{S} \times \mathcal{Z}$ and action $a \in \mathcal{A}$. Since z is sampled from a prior p_z which is independent of state, the transition dynamics $\hat{p}$ and reward function $\hat{r}$ are given as:

$$\hat{p}(\hat{s}' | \hat{s}, a) = p(s' | s, a) p_z(z'), \quad \hat{r}(\hat{s}, a) = r(s, a). \quad (10)$$

We hereafter refer to this construction as the **z-MDP**, and adopt the notation $(\cdot)$ to distinguish entities in the z-MDP from those in the original MDP. In Corollary 4.2, we show that any expectation over a measurable function is unchanged whether you average over $\hat{\rho}_\pi(\hat{s}, z)$ or $\rho_\pi(s, z)$.

Corollary 4.2 (Marginal consistency). Given $\hat{\pi}$ and π in Section 4.1, the discounted state-action marginal distribution $\hat{\rho}_\pi(s, z, a)$ in the z-MDP $\hat{\mathcal{M}}$ satisfies $\rho_\pi(s, a) = \int \hat{\rho}_\pi(s, z, a) dz$. Consequently, the following equation holds for any measurable function $f : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$

$$\mathbb{E}_{(s, z, a) \sim \hat{\rho}_\pi} [f(s, a)] = \mathbb{E}_{(s, a) \sim \rho_\pi} [f(s, a)]. \quad (11)$$

Cross-entropy. The MaxEnt-RL objective requires the intractable entropy of π , as the marginalization over z in Eq. (9) has no closed form. To circumvent this, we propose using the cross-entropy between π and $\tilde{\pi}_\theta$, $\mathcal{H}(\pi(\cdot | s), \tilde{\pi}_\theta(\cdot | s)) \triangleq \mathbb{E}_{a \sim \pi(\cdot | s)} [-\log \tilde{\pi}_\theta(a | s)]$, which can be evaluated in an unbiased manner via the Hutchinson trace estimator (see Appendix A for details). The cross-entropy is reformulated as an expectation over z :

$$\mathcal{H}(\pi(\cdot | s), \tilde{\pi}_\theta(\cdot | s)) = \mathbb{E}_{z \sim p_z} [\mathcal{H}(\hat{\pi}(\cdot | \hat{s}), \tilde{\pi}_\theta(\cdot | s))] \quad (12)$$

For brevity, we denote $\tilde{\pi}_\theta$ by $\tilde{\pi}$ throughout this section. The following proposition establishes that the cross-entropy converges to the entropy of π .

Proposition 4.3 (Cross-Entropy as a Valid Entropy Surrogate). *For smooth $\tilde{\pi}$ and $\text{tr}(\Sigma) \ll 1$, the local variance governs the Total Variation (TV) distance and KL divergence between π and $\tilde{\pi}$:*

$$D_{\text{TV}}(\pi, \tilde{\pi}) = \mathcal{O}(\sqrt{\text{tr}(\Sigma)}), \quad D_{\text{KL}}(\pi \parallel \tilde{\pi}) = \mathcal{O}(\text{tr}(\Sigma)^2). \quad (13)$$

Consequently, as the local variance vanishes ($\text{tr}(\Sigma) \rightarrow 0$), the surrogate cross-entropy converges to the true entropy of the global policy:

$$\mathcal{H}(\pi, \tilde{\pi}) = \mathcal{H}(\pi) + \mathcal{O}(\text{tr}(\Sigma)^2) \approx \mathcal{H}(\pi). \quad (14)$$

Proposition 4.3 justifies a tractable proxy for the MaxEnt-RL objective in $\mathcal{M}$ via the cross-entropy under a small-variance assumption on the local Gaussian.

Proxy MaxEnt-RL Objective. Using the cross-entropy, we now define the proxy MaxEnt-RL objective and corresponding soft value functions. The augmented policy objective is given as

$$\mathcal{J}(\pi) = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{(s_t, a_t) \sim \rho_{\pi}} [(r(s_t, a_t) - \alpha \log \tilde{\pi}(a_t | s_t))]. \quad (15)$$

By Corollary 4.2 and Eq. (12), this objective is equivalently evaluated in the z -MDP as $\mathcal{J}(\hat{\pi})$, providing a theoretical bridge for optimization within the latent-augmented space $\hat{\mathcal{M}}$.

Value function. Following SAC [20], we define the soft state-value function and the corresponding soft Bellman operator $\mathcal{T}^{\pi}$ by replacing the entropy term with the cross-entropy term:

$$\text{(Soft Value Function)} \quad V^{\pi}(s) = \mathbb{E}_{a \sim \pi} [Q^{\pi}(s, a) - \alpha \log \tilde{\pi}(a | s)] \quad (16)$$

$$\text{(Soft Bellman Operator)} \quad (\mathcal{T}^{\pi} Q)(s, a) \triangleq r(s, a) + \gamma \mathbb{E}_{s' \sim p} [V^{\pi}(s')] \quad (17)$$

In the z -MDP $\hat{\mathcal{M}}$, the Q-function is defined with $Q^{\hat{\pi}}(\hat{s}, a)$.

Corollary 4.4 (Q-function consistency). *By Corollary 4.2, we use the same Q-function both in $\mathcal{M}$ and $\hat{\mathcal{M}}$*

$$Q^{\hat{\pi}}(\hat{s}, a) = Q^{\pi}(s, a), \quad \forall s, z, a \quad (18)$$

Therefore, we use the Q-function $Q^{\pi}(s, a)$ of the original MDP in z -MDP.

According to Corollaries 4.2 and 4.4, maximizing the Q-function $Q^{\pi}(s, a)$ via the local policy $\hat{\pi}$ for all $\hat{s} \sim \hat{\rho}_{\hat{\pi}}$ in the z -MDP is equivalent to optimizing the global policy π for all $s \sim \rho_{\pi}$.

4.3 FLAG: A Practical Local Policy Update Algorithm

In this section, we extend the EM updates from Section 3.1 to the z -MDP and our local policy $\hat{\pi}(\cdot | \hat{s}; \theta)$. At each iteration k , we treat $\tilde{\pi}_{\theta_k}$ as a fixed reference, making the augmented reward $r_t - \alpha \log \tilde{\pi}_{\theta_k}(a_t | s_t)$ play the role of an iteration-fixed reward, enabling the EM derivation

$$\mathcal{J}_{\text{EM}}(q, \theta) = \mathbb{E}_{\hat{\tau} \sim p_q} \left[\sum_{t=0}^{\infty} \gamma^t ((r_t - \alpha \log \tilde{\pi}_{\theta_k}(a_t | s_t)) / \lambda) \right] - D_{\text{KL}}(p_q(\hat{\tau}) \parallel p_{\hat{\pi}}(\hat{\tau})). \quad (19)$$

Crucially, the cross-entropy structure allows us to define and evaluate this objective for a variational distribution q , which admits E-step target in a closed form. For a reference policy $\hat{\pi}_k$ parameterized by θ_k and its associated Q-function Q^{π_k} , we define the energy function for brevity

$$f_{\hat{s}, k}(a) := Q^{\pi_k}(s, a) - \alpha \log \tilde{\pi}_{\theta_k}(a | s). \quad (20)$$

For each $\hat{s} \sim \hat{\rho}_{\hat{\pi}_k}$,

$$\text{(E-step:)} \quad q_k(a | \hat{s}) \propto \hat{\pi}(a | \hat{s}; \theta_k) \exp(f_{\hat{s}, k}(a) / \lambda) \quad (21)$$

$$\text{(M-step:)} \quad \theta_{k+1} \in \arg \max_{\theta} \mathbb{E}_{a \sim q_k} [\log \hat{\pi}(a | \hat{s}; \theta)] \quad (22)$$

M-step via Moment Matching. Since $\hat{\pi}$ is Gaussian with fixed Σ_k , the M-step in Eq. (22) reduces to matching the first moment :

$$\theta_{k+1} \in \arg \min_{\theta} \mathbb{E}_{(s, z) \sim \hat{\rho}_{\hat{\pi}}} \|T_{\theta}(s, z) - \mu_k^*(\hat{s})\|^2, \quad \mu_k^*(\hat{s}) := \mathbb{E}_{q_k}[a]. \quad (23)$$

We approximate μ_k^* via self-normalized importance sampling [8] with N samples $\{\delta_i\}_{i=1}^N \sim \hat{\pi}_k(\cdot | \hat{s})$:

$$\mu_k^*(\hat{s}) = \mathbb{E}_{q_k}[a] \approx \sum_{i=1}^N \bar{w}_i a_i, \quad \bar{w}_i = \text{softmax}(f_{\hat{s}, k}(a + \delta_i) / \lambda). \quad (24)$$

We treat $\mu_k^*(\hat{s})$ as an improved action label for the anchor action $T_{\theta_k}(s, z)$. To realize Eq. (23), we distill these labels into the base flow policy $\tilde{\pi}_{\theta_k}$ using the CFM objective in Eq. (7), i.e.,

$$\mathcal{L}(\theta; s, z) = \mathbb{E}_{\tau} [\|u_{\theta}^{\tau}(s, \zeta^{\tau}(z, \mu^*)) - (\mu^* - z)\|^2]. \quad (25)$$

This label acts as a *guiding flag*, as it indicates the latent vector z where to generate in the action space (Figure 2).

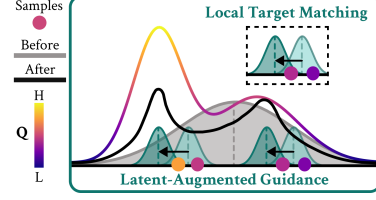


Figure 2: Target matching is performed on the local policy and target actions are distilled back to the flow policy.

Monotonic improvement guarantee. We now show that FLAG’s EM update monotonically improves the variational objective, $\mathcal{J}_{k+1} \geq \mathcal{J}_k$, where $\mathcal{J}_k := \mathcal{J}_{\text{EM}}(q_k, \theta_k)$. Let $\mathcal{G}_k$ denote the squared gradient norm of the Eq. (23) and the local covariance be $\Sigma_k = \sigma_k^2 I$.

Theorem 4.5 (Monotonic Improvement Guarantee). *Let the CFM update (Eq. (25)) approximate the ideal KL projection (Eq. (23)) up to an error ϵ_k^{proj} . For sufficiently small step size β the one-step improvement of the FLAG update is lower-bounded by*

$$\mathcal{J}_{k+1} - \mathcal{J}_k \geq \underbrace{\lambda \beta \mathcal{G}_k}_{\text{frozen-reward MPO improvement}} - \underbrace{\alpha C_{\Sigma} \sigma_k^2 \beta \mathcal{G}_k}_{\text{cross-entropy reward drift}} - \underbrace{\lambda \epsilon_k^{\text{proj}}}_{\text{CFM projection error}} + \mathcal{O}(\beta^2), \quad (26)$$

where $C_{\Sigma} > 0$ is a constant absorbing zeroth-order approximation residuals. Consequently, whenever $\lambda > \alpha C_{\Sigma} \sigma_k^2$ and ϵ_k^{proj} is sufficiently small, $\mathcal{J}_{k+1} \geq \mathcal{J}_k$.

The three terms in Theorem 4.5 correspond to the standard MPO improvement under frozen reward, the drift of the cross-entropy reward as θ_k changes, and the approximation error introduced by the CFM realization of the KL projection (see Figure 6 for the proof overview). We ensure the sufficient conditions $\lambda > \alpha C_{\Sigma} \sigma_k^2$ and $\epsilon_k^{\text{proj}} \rightarrow 0$ are satisfied through the design choices in Section 4.4. While Theorem 4.5 follows variational framework, Figure 5 and Appendix C.2 provide a complementary derivation showing how FLAG’s moment-matching update recovers SAC’s soft policy improvement.

4.4 Implementation Details

Autotuning Temperature. We follow [20] to automatically tune the entropy scaling term α :

$$\mathcal{L}(\alpha) = \alpha(\mathcal{H}(\pi, \tilde{\pi}) - \mathcal{H}_{\text{target}}), \quad (27)$$

where $\mathcal{H}_{\text{target}}$ is a hyperparameter that determines the stochasticity of the policy.

Effective Temperature. We normalize the energy function as $f_{\text{norm}}(a) = f_{\hat{s},k}(a)/\alpha$ with a fixed reference temperature λ_{ref} [17, 22, 26], equivalent to using an effective temperature $\lambda = \alpha \lambda_{\text{ref}}$ in Eq. (21). Fixing λ_{ref} avoids the numerical instability of optimizing two Lagrange multipliers α and λ , and reduces the sufficient condition for monotonic improvement (Theorem 4.5) to $\lambda_{\text{ref}} > C_{\Sigma} \sigma_k^2$.

Covariance Scheduling. We use an isotropic covariance $\Sigma_k = \sigma_k^2 I$. σ_k is annealed from σ_{init} to a small σ_{final} , keeping σ_k small throughout training. This scheduling is crucial in Theorem 4.5: since the reward-drift term scales as $\mathcal{O}(\sigma_k^2)$, annealing $\sigma_k \rightarrow 0$ tightens the guarantee.

Guidance Buffer. We cache locally improved action labels $\mu_k^*(\hat{s})$ in a small guidance buffer and reuse them across multiple off-policy updates [39, 44]. Since the E-step targets drift slowly across iterations, recent labels remain effective targets for the current policy update, providing dense supervision that keeps the CFM projection error ϵ_k^{proj} in Theorem 4.5 small in practice.

Other Details. Following DIME [11], we employ a distributional critic [5] trained via CrossQ [7]. Full implementation details and the training procedure are provided in Appendix D and Algorithm 1.

FLAG realizes local IS through EM on the z -MDP.

The z -MDP augments the original MDP with a latent variable while preserving marginal and Q -function consistency. On this augmented MDP, we define a tractable proxy MaxEnt-RL objective whose EM-style optimization induces a localized proposal-target pair conditioned on z . FLAG instantiates this procedure, enabling the local IS within the localized region of the action space and making MaxEnt-RL scalable.

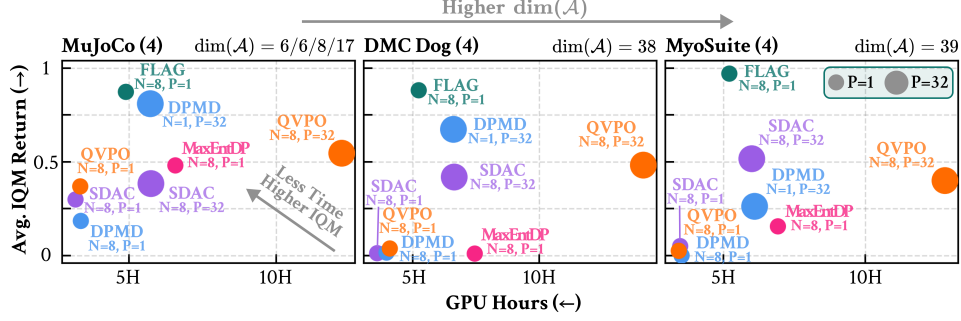


Figure 3: **Performance and computational efficiency of global and local proposal sampling.** We plot normalized performance against wall-clock GPU runtime to examine the performance–efficiency trade-off. The x -axis denotes the time (hours) for a single 1M-step training run on an NVIDIA L40S GPU, and the y -axis reports aggregated IQM returns across 10 random seeds at 1M steps. A method in the *upper-left* region is both higher-performing and takes less time to train. P denotes the number of samples in the best-of- P heuristic, and is indicated by the diameter of each marker. FLAG consistently occupies the upper-left region across all domains, demonstrating superior performance *without* additional computational overhead.

5 Experimental Results

We answer three questions about FLAG in this section.

- (Q1) Section 5.1: Does FLAG scale to high-dimensional action spaces under limited sample budgets?
- (Q2) Section 5.2: How does FLAG compare to action-gradient and BPTT-based actor-critic methods?
- (Q3) Section 5.3: How do our key design choices—covariance scheduling and the guidance buffer—connect to the theoretical results?

Ablations on the variance reduction of the Hutchinson trace estimator, the effect of cross-entropy, variations on λ_{ref} , and the number of ODE steps are deferred to Appendix F. Hyperparameters and experiment details are deferred to Appendix G.

Benchmarks and Metrics. We evaluate online RL algorithms on three challenging benchmark suites: DMC suite [46], MyoSuite [10], and MuJoCo [43]. MuJoCo is included because target-matching policy optimization methods are commonly evaluated in this domain. We evaluate on four tasks per benchmark: HalfCheetah, Walker2d, Ant, and Humanoid for MuJoCo-v5; Dog-Stand, Dog-Walk, Dog-Trot, and Dog-Run for DMC Dog; and Reach-Hard, Obj-Hold-Hard, Key-Turn-Hard, and Pen-Twirl-Hard for MyoSuite. To compare results across benchmarks with different reward scales, we report normalized scores: MuJoCo scores are normalized by CrossQ performance [7], DMC Dog returns are divided by the maximum return of 1,000, and MyoSuite scores are reported as success rates.

5.1 FLAG is Scalable to High-dimensional Action Spaces and Robust to Sample Budget

Is FLAG scalable to high-dimensional action space?

To answer the question, we first compare FLAG with the global IS baselines: SDAC and DPMD [27], QVPO [13], and MaxEntDP [16]. This comparison highlights the limited scalability of the global IS in high-dimensional action spaces under the same conditions. QVPO is included as a baseline since it also obtains samples from global proposal distribution to estimate a lower bound on the policy gradient. All methods use the same CrossQ [7] critic with distributional critics [5], and the number of Q-function evaluations per policy update (N) is fixed to 8 due to computational cost. Our main comparison sets the best-of- P budget to $P=1$, where best-of- P selects the highest- Q action among P sampled candidates, to isolate the effect of the proposal distribution. We additionally report $P=32$ variants where applicable.¹

As shown in Figure 3, global-proposal baselines struggle in high-dimensional action spaces, especially without best-of- P . Even with the stronger $P=32$ heuristic, their performance remains substantially

¹The $P=32$ variant is not applied to MaxEntDP because its policy-update candidates are drawn from the replay buffer, making best-of- P selection inapplicable.

below FLAG. In contrast, FLAG achieves consistently higher returns with the same Q-function evaluation budget ($N = 8$), indicating that local proposal matching is the key mechanism enabling scalable target matching in high-dimensional control. We also present the learning curves in Figure 9 and Figure 10.

To rule out the possibility that FLAG’s advantage comes from the use of CrossQ or from an artificially restricted Q-function inference budget, we evaluate methods without CrossQ. In this setting, the baselines use their default, larger sample budgets. As shown in Figure 4, the global-proposal baselines still fail to learn meaningful behaviors on challenging DMC Dog tasks, whereas FLAG remains effective. This suggests that the performance gap is not explained by the value-learning architecture or the evaluation budget, but by the proposed local proposal matching strategy.

Is FLAG robust to the sample budget?

We evaluate the sensitivity of FLAG to the sample budget. As shown in Table 1, FLAG remains robust even with limited samples ($N = 2$), exhibiting only mild performance degradation under limited sampling budgets. This robustness provides empirical evidence for our main design principle: by localizing both the proposal and target distributions, FLAG reduces their discrepancy and makes importance sampling effective within a restricted region of the action space. In contrast to global IS, whose samples are spread over the full action space and may have little overlap with the target density, local IS operates between two distributions supported on the same local region. As a result, the importance weights remain informative even with few samples.

Table 1: Ablation study on the number of training samples in DMC Dog tasks. We aggregate 10 random seeds at 1M steps. The **highest** and **second** scores are highlighted.

Num. Samples	Dog-trot Return (1k)	Dog-run Return (1k)
$N = 2$	0.620 ± 0.072	0.818 ± 0.095
$N = 4$	0.639 ± 0.061	0.827 ± 0.106
$N = 8$	0.680 ± 0.066	0.912 ± 0.047
$N = 16$	0.687 ± 0.073	0.874 ± 0.087
$N = 32$	0.641 ± 0.075	0.879 ± 0.057

5.2 FLAG Outperforms BPTT-based Actor-Critic and Action Gradient Methods

FLAG does not rely on critic gradients, making it robust even when critic gradient signals are unreliable. To investigate this, we compare methods in two settings: with and without CrossQ. For BPTT baselines, we use DIME [11], FlowRL [26], and DACERv2 [48], and for action-gradient baselines, we use DIPO [49] and QSM [38]. Following Section 5.1, we match critics using CrossQ and distributional critics across all methods². As shown in Table 2, FLAG outperforms all baselines across high-dimensional benchmarks. To further examine FLAG’s robustness to critic quality, we conduct experiments without CrossQ (Figure 4)³. While gradient-based methods are susceptible to noisy signals from an insufficiently trained critic, FLAG aggregates critic signals through importance-weighted averaging rather than direct differentiation, smoothing out noise in the learning target and thereby maintaining stable performance.

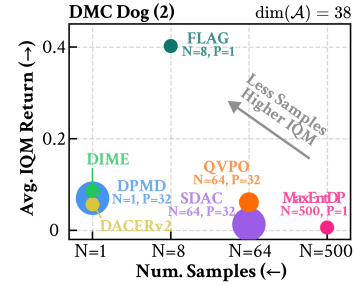


Figure 4: Comparison on the DMC Dog-run and Dog-trot task w/o CrossQ, performance averaged. We report with 10 random seeds for $P = 1$ and 5 for $P = 32$.

5.3 FLAG Key Design Choices Align with Theoretical Results

We empirically show that the key design choices described in Section 4.4 help control the terms in the monotonic improvement bound of Theorem 4.5. This bound identifies two controllable quantities in practice: (i) the covariance-dependent drift term and (ii) the CFM projection error ϵ_k^{proj} . Accordingly, FLAG uses covariance scheduling and a guidance buffer (Section 4.4). In this section, we ablate these two components to examine how they affect the corresponding terms in practice.

Covariance Scheduling The condition $\lambda > \alpha C_{\Sigma} \sigma_k^2$ becomes easier to satisfy as the local covariance decreases, since σ_k^2 directly suppresses the reward-drift term (Eq. (26)). However, an overly small σ_k can restrict the proposal region, thus weaken the guidance signal. To balance these effects, FLAG anneals the local covariance from σ_{init} to σ_{final} . As shown in Table 3, moderate annealing

²Since DACERv2 trains critic to predict the mean and standard deviation of the Q-value, we exclude distributional critics.

³We only include DACERv2 and DIME in this comparison, as these methods achieve comparable performance in Table 2.

Table 2: IQM final performance after 1M environment steps on DMC Dog and MyoSuite. All results are computed from evaluation collected over 10 seeds with 5 evaluation episodes. Each entry reports the IQM point estimate with a 95% confidence interval; the $\pm$ notation is used only as a compact summary of the confidence interval ([23]). The **highest** and **second** IQM point estimates are highlighted in each column, respectively.

DMC Dog (4)							
Policy	Method	Alg.	Dog-stand	Dog-walk	Dog-trot	Dog-run	Avg.
Gauss.	-	CrossQ	0.957 $\pm$ 0.121	0.882 $\pm$ 0.079	0.890 $\pm$ 0.016	0.467 $\pm$ 0.080	0.799
	Action Gradient	DIPO QSM	0.906 $\pm$ 0.030 0.061 $\pm$ 0.131	0.570 $\pm$ 0.164 0.137 $\pm$ 0.144	0.404 $\pm$ 0.080 0.142 $\pm$ 0.053	0.264 $\pm$ 0.017 0.064 $\pm$ 0.055	0.536 0.101
Express.	BPTT-based Actor-Critic	DACERv2	0.923 $\pm$ 0.031	0.726 $\pm$ 0.258	0.367 $\pm$ 0.226	0.090 $\pm$ 0.108	0.526
		DIME	0.968 $\pm$ 0.017	0.944 $\pm$ 0.024	0.886 $\pm$ 0.050	0.629 $\pm$ 0.056	0.857
		FlowRL	0.964 $\pm$ 0.014	0.910 $\pm$ 0.011	0.870 $\pm$ 0.018	0.452 $\pm$ 0.036	0.799
	Target Matching	FLAG (Ours)	0.974 $\pm$ 0.016	0.950 $\pm$ 0.011	0.912 $\pm$ 0.044	0.680 $\pm$ 0.062	0.879
MYOSUITE (4)							
Policy	Method	Alg.	Reach-Hard	Obj-Hold-Hard	Key-Turn-Hard	Pen-Twirl-Hard	Avg.
Gauss.	-	CrossQ	0.600 $\pm$ 0.217	1.000 $\pm$ 0.100	0.900 $\pm$ 0.150	0.100 $\pm$ 0.133	0.650
	Action Gradient	DIPO QSM	0.600 $\pm$ 0.219 0.089 $\pm$ 0.069	0.340 $\pm$ 0.201 0.000 $\pm$ 0.000	0.560 $\pm$ 0.408 0.209 $\pm$ 0.153	0.760 $\pm$ 0.174 0.026 $\pm$ 0.048	0.565 0.081
Express.	BPTT-based Actor-Critic	DACERv2	0.800 $\pm$ 0.107	0.925 $\pm$ 0.157	0.911 $\pm$ 0.100	0.800 $\pm$ 0.129	0.859
		DIME	0.900 $\pm$ 0.167	1.000 $\pm$ 0.017	1.000 $\pm$ 0.000	0.700 $\pm$ 0.233	0.900
		FlowRL	0.000 $\pm$ 0.008	0.017 $\pm$ 0.042	0.000 $\pm$ 0.000	0.050 $\pm$ 0.067	0.017
	Target Matching	FLAG (Ours)	0.930 $\pm$ 0.078	1.000 $\pm$ 0.000	0.930 $\pm$ 0.142	0.870 $\pm$ 0.174	0.933

Table 3: Ablation study on buffer size and covariance scheduling. We report returns normalized by 1k in DMC Dog-run tasks. $\dagger$ denotes the hyperparameters used in Section 5.1 and Section 5.2. All experiments are conducted with 5 random seeds. The **highest** and **second** scores are highlighted in each column, respectively.

	Buffer Size					Covariance Scheduling σ_{init} (σ_{final})							
	0	10.24k †	51.2k	102.4k	204.8k	-1(-1)	-1(-2)	-1(-3)	-2(-2)	-2(-3) †	-2(-4)	-2(-5)	-2(-6)
Return (1k) $\uparrow$	0.601	0.680	0.670	0.618	0.589	0.614	0.675	0.732	0.588	0.680	0.597	0.547	0.269

287 yields improved performance by allowing early local exploration while gradually reducing the drift. In
 288 contrast, overly aggressive annealing shrinks the local search region too early, leading to convergence
 289 to suboptimal local modes and eventual policy collapse.

290 **Guidance Buffer** The CFM projection error ϵ_k^{proj} measures how accurately the flow policy realizes
 291 the improved target actions obtained from the M-step. The guidance buffer reduces this error by
 292 caching recently computed target actions and reusing them across multiple off-policy updates, thereby
 293 providing denser supervision for the CFM projection. As shown in Table 3, removing the guidance
 294 buffer degrades performance, suggesting insufficient projection of local improvements into the flow
 295 policy. Conversely, an excessively large buffer also hurts performance, as stale targets become
 296 increasingly mismatched with the current policy. A moderate buffer size achieves the best trade-off
 297 between supervision density and target freshness, keeping ϵ_k^{proj} small in practice.

298 6 Conclusion

299 We presented FLAG, a MaxEnt-RL framework that optimizes an expressive policy through a super-
 300 vised learning paradigm via latent-augmented guidance. By leveraging the deterministic property
 301 of the flow map, FLAG replaces the global proposal distribution of prior importance sampling-
 302 based methods with a local one, avoiding both the weight degeneracy of global sampling and the
 303 numerical instability of BPTT. Empirically, FLAG scales to high-dimensional action spaces with
 304 limited Q-function evaluations, and performs competitively with methods that rely on BPTT and
 305 critic gradients. To the best of our knowledge, this is the first supervision-based approach to scale
 306 to such high-dimensional control environments. As a limitation, the combination of the base flow
 307 policy and local Gaussian head is one particular instantiation of our framework; a more principled
 308 construction may be obtained via ODE-to-SDE conversion [15], which we leave for future work.

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A Computational Challenges in Flow-based MaxEnt-RL

In this section, we describe in detail the technical challenges of applying flow-based policies within the standard MaxEnt-RL framework. Specifically, we define our policy $\tilde{\pi}_\theta$ as a flow-induced distribution where an action is generated by solving the ODE $T_\theta(s, z) = a^1 = z + \int_0^1 v_\theta(a^\tau, \tau, s) d\tau$ following Eq. (6).

Unbiased log-density estimation using Hutchinson trace estimator. A significant advantage of flow-based policies is the ability to recover exact log densities, which are typically unavailable in diffusion models. For a flow policy defined by the ODE trajectory ϕ_t^τ , the log-density admits a path-integral form:

$$\log \tilde{\pi}_\theta(a_t | s_t) = \log p_z(z_t) - \int_0^1 \text{tr}(J_\theta^\tau) d\tau, \quad J_\theta^\tau \triangleq \frac{\partial u_\theta^\tau}{\partial \phi}(s_t, \phi_t^\tau).$$

Hutchinson’s identity [21] allows us to estimate the trace of a square matrix J . Using any random vector ϵ s.t. $\mathbb{E}[\epsilon \epsilon^\top] = I$, we obtain the unbiased estimator, $\text{Tr}(J) = \mathbb{E}_\epsilon[\epsilon^\top J \epsilon]$. Applying this estimator to $\log \tilde{\pi}_\theta(a | s)$ yields the following.

$$\log \tilde{\pi}_\theta(a_t | s_t) = \mathbb{E}_\epsilon \left[\log p_z(z_t) - \int_0^1 \epsilon^\top J_\theta^\tau \epsilon d\tau \right]. \quad (28)$$

In practice, the quadratic form $\epsilon^\top J_\theta^\tau \epsilon$ is computed without forming J_θ^τ explicitly leveraging Jacobian-vector product. With reparameterization $a_t = T_\theta(s_t, z_t)$ and Eq. (28), we can form an unbiased Monte Carlo estimator of MaxEnt objective as follows:

$$\mathcal{J}_{\text{MaxEnt}}(\theta) = \mathbb{E}_{z_t \sim p_z, \epsilon} \left[Q_\phi(s_t, T_\theta(s_t, z_t)) - \log p_z(z_t) - \int_0^1 \epsilon^\top J_\theta^\tau \epsilon d\tau \right]. \quad (29)$$

Pathwise gradient requires differentiating through the entire ODE. Although Eq. (29) enables unbiased *evaluation*, optimizing θ by a direct pathwise gradient requires $\partial a_t / \partial \theta$. Let $S_t^\tau \triangleq \frac{\partial \phi_t^\tau}{\partial \theta}$ denote the parameter sensitivity along the flow. Differentiating the ODE gives the sensitivity equation

$$\frac{dS_t^\tau}{d\tau} = J_\theta^\tau S_t^\tau + \frac{\partial u_\theta^\tau}{\partial \theta}(s_t, \phi_t^\tau), \quad S_t^0 = 0, \quad \frac{\partial a_t}{\partial \theta} = S_t^1. \quad (30)$$

Therefore, the pathwise gradient of the Q -term is

$$\nabla_\theta Q_\phi(s_t, a_t) = \nabla_a Q_\phi(s_t, a) \big|_{a=a_t} S_t^1. \quad (31)$$

Computing S_t^1 entails differentiating through the entire ODE evolution in $\tau \in [0, 1]$. Since the sensitivity dynamics repeatedly apply the local Jacobian J_θ^τ along the trajectory, the accumulated transformation can become ill-conditioned, leading to vanishing/exploding sensitivities [6, 34].

The log-density gradient involves second-order terms (HVP). The Hutchinson integrand depends on the state through $J_\theta^\tau = \partial u_\theta^\tau / \partial \phi$. Its state gradient is

$$\nabla_{\phi_t^\tau} (\epsilon^\top J_\theta^\tau \epsilon) = \left\langle \epsilon, \left(\nabla_\phi^2 u_\theta^\tau(s_t, \phi_t^\tau) \right) \epsilon \right\rangle,$$

which is a Hessian–vector product (a double contraction with ϵ). Propagating this along the flow yields the score term

$$\nabla_a \log \tilde{\pi}_\theta(a_t | s_t) = J_0^\top \nabla_z \log p_z(z_t) - \mathbb{E}_\epsilon \left[\int_0^1 (J_\theta^\tau)^\top \underbrace{\left\langle \epsilon, \left(\nabla_\phi^2 u_\theta^\tau(s_t, \phi_t^\tau) \right) \epsilon \right\rangle}_{\text{HVP}} d\tau \right], \quad (32)$$

where $\nabla_a = (J_\theta^\tau)^\top \nabla_{\phi_t^\tau}$.

Implication. Eq. (30) and Eq. (31) show that even the Q -term requires differentiating through the entire ODE. More importantly, Eq. (32) reveals that the entropy gradient additionally involves Hessian–vector products of the vector field, making direct MaxEnt-RL optimization with flow policies more expensive than objective evaluation.

B Derivations

In this section, we make detail derivations in Section 4.

B.1 Global and Local Consistency

B.1.1 Marginal Distributions Consistency (Corollary 4.2)

While our general framework assumes an infinite horizon, for notational clarity, we restrict our attention to a finite horizon T in this derivation. Fix an initial pair (s_t, a_t) , and any z_t ; write $\hat{s}_t = (s_t, z_t)$. Consider the suffix trajectories

$$\tau = (s_{t+1}, a_{t+1}, \dots, s_T, a_T), \quad \hat{\tau} = (\hat{s}_{t+1}, a_{t+1}, \dots, \hat{s}_T, a_T).$$

Under the augmented process with z -conditioned policy $\hat{\pi}(\cdot | \hat{s})$ and i.i.d. z 's,

$$p_{\hat{\pi}}(\hat{\tau} | \hat{s}_t, a_t) = \prod_{i=t}^{T-1} p(s_{i+1} | s_i, a_i) p_z(z_{i+1}) \hat{\pi}(a_{i+1} | s_{i+1}, z_{i+1}). \quad (33)$$

Marginalizing out $z_{t+1:T}$ gives the (s, a) -trajectory law

$$\begin{aligned} \int p_{\hat{\pi}}(\hat{\tau} | \hat{s}_t, a_t) \prod_{i=t+1}^{T-1} dz_i &= \prod_{i=t}^{T-1} p(s_{i+1} | s_i, a_i) \underbrace{\int p_z(z_{i+1}) \hat{\pi}(a_{i+1} | s_{i+1}, z_{i+1}) dz_{i+1}}_{= \pi(a_{i+1} | s_{i+1})} \\ &= \prod_{i=t}^{T-1} p(s_{i+1} | s_i, a_i) \pi(a_{i+1} | s_{i+1}) =: p_{\pi}(\tau | s_t, a_t). \end{aligned} \quad (34)$$

Thus, after integrating out z , the (s, a) -trajectory distribution exactly matches that induced by the marginal policy π . We can extend this result to the infinite horizon as $T \rightarrow \infty$.

Lemma B.1. *In the z -MDP framework, let $\hat{\pi}$ be the local policy defined in Eq. (8) and π be the global policy defined in Eq. (9). The discounted state and state-action marginal distributions induced by $\hat{\pi}$ satisfy:*

$$\hat{\rho}_{\hat{\pi}}(s, z, a) = \rho_{\pi}(s) p_z(z) \hat{\pi}(a | s, z), \quad \hat{\rho}_{\hat{\pi}}(\hat{s}) = \rho_{\pi}(s) p_z(z). \quad (35)$$

Proof. Consider the joint probability of the state and latent variable at time step t . Since z_t is sampled i.i.d. from $p_z(\cdot)$ and is independent of s_t , we have $P_{\hat{\pi}}(z_t = z | s_t = s) = p_z(z)$. Furthermore, leveraging the trajectory consistency in Eq. (34), the marginal distribution of s_t under $\hat{\pi}$ is identical to that under π , i.e., $P_{\hat{\pi}}(s_t = s) = P_{\pi}(s_t = s)$.

Combining these observations, the joint probability at time t factorizes as:

$$P_{\hat{\pi}}(s_t = s, z_t = z) = P_{\pi}(s_t = s) p_z(z). \quad (36)$$

Substituting this equality into the definition of the discounted state marginal distribution yields:

$$\begin{aligned} \hat{\rho}_{\hat{\pi}}(s, z) &= (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t P_{\hat{\pi}}(s_t = s, z_t = z) \\ &= (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t P_{\pi}(s_t = s) p_z(z) \\ &= \left[(1 - \gamma) \sum_{t=0}^{\infty} \gamma^t P_{\pi}(s_t = s) \right] p_z(z) \\ &= \rho_{\pi}(s) p_z(z). \end{aligned} \quad (37)$$

Finally, for the state-action marginal distribution, the following equation concludes the proof:

$$\hat{\rho}_{\hat{\pi}}(s, z, a) = [\rho_{\pi}(s) p_z(z)] \hat{\pi}(a | s, z), \quad \rho_{\pi}(s, a) = \int p_z(z) \hat{\pi}(a | s, z) \rho_{\pi}(s) dz. \quad (38)$$

□

493 B.1.2 Q-function Consistency (Corollary 4.4)

494 Now we compare soft returns. The augmented definition is

$$Q^{\hat{\pi}}(\hat{s}_t, a_t) = \mathbb{E}_{\hat{\tau} \sim p_{\hat{\pi}}} \left[\sum_{k=0}^{\infty} \gamma^k (\hat{r}(\hat{s}_{t+k}, a_{t+k}) + \alpha \mathcal{H}(\hat{\pi}(\cdot | \hat{s}_{t+k}), \tilde{\pi}(\cdot | s_{t+k}))) \right].$$

495 Taking expectation over $z_{t+1:\infty}$ and using Eq. (34), the reward terms coincide by $\hat{r}(\hat{s}, a) = r(s, a)$,
496 and the cross-entropy terms reduce to

$$\mathbb{E}_{z_{t+k} \sim p_z} [H(\hat{\pi}(\cdot | \hat{s}_{t+k}), \tilde{\pi}(\cdot | s_{t+k}))] = \mathcal{H}(\pi(\cdot | s_{t+k}), \tilde{\pi}(a | s_{t+k})).$$

497 Hence, marginalizing over $z_{t+1:\infty}$ gives

$$Q^{\pi}(\hat{s}_t, a_t) = \mathbb{E}_{\tau \sim p_{\pi}} \left[\sum_{k=0}^{\infty} \gamma^k \left(r(s_{t+k}, a_{t+k}) + \alpha \mathcal{H}(\pi(\cdot | s_{t+k}), \tilde{\pi}(a | s_{t+k})) \right) \right] =: Q^{\pi}(s_t, a_t).$$

498 The right-hand side does not depend on the initial z_t , therefore $Q^{\hat{\pi}}(s_t, z_t, a_t) = Q^{\pi}(s_t, a_t)$ is valid
499 for all z_t .

500 B.2 EM algorithm Derivations (Section 4.3)

501 This section provides the full derivation of FLAG’s EM update in Section 4.3. We proceed in three
502 steps: (i) we cast cross-entropy augmented RL objective as probabilistic inference and derive the
503 variational lower bound both in the original MDP and the z -MDP; (ii) we solve the E-step in closed
504 form, yielding a non-parametric target distribution q_k ; (iii) we project q_k back to the parametric policy
505 class through the M-step. Throughout, we follow the MPO [2] framework while adapting it to our
506 cross-entropy augmented reward structure, which we discuss in the remark below.

507 An infinite-horizon discounted reward formulation can be cast as inference problem as follows:

$$p(\mathcal{O} = 1 | \tau) \propto \exp \left(\sum_{t=1}^{\infty} \gamma^t (r_t / \lambda) \right) \quad (39)$$

508 where $r_t = r(s_t, a_t)$. Here, the optimality variable $\mathcal{O}$ reveals *the event of obtaining maximum reward*
509 *by choosing an action*. We define the augmented reward in the spirit of Haarnoja et al. [20].

$$\begin{aligned} r_{\pi}(s_t, a_t) &\triangleq r(s_t, a_t) + \alpha \mathbb{E}_{s_{t+1} \sim p(\cdot | s_t, a_t)} [\mathcal{H}(\pi(\cdot | s_{t+1}), \tilde{\pi}(\cdot | s_{t+1}))], \\ \hat{r}_{\hat{\pi}}(\hat{s}_t, a_t) &\triangleq r(\hat{s}_t, a_t) + \alpha \mathbb{E}_{\hat{s}_{t+1} \sim \hat{p}(\cdot | \hat{s}_t, a_t)} [\mathcal{H}(\hat{\pi}(\cdot | \hat{s}_{t+1}), \tilde{\pi}(\cdot | s_{t+1}))]. \end{aligned} \quad (40)$$

Remark. Remark on the reward structure and the E-step. The augmented reward in Eq. (40) is defined separately for each policy (r_{π} for π , r_q for q), since the cross-entropy bonus follows the soft Bellman structure of SAC [20]. Although this augmentation is policy-dependent, it is compatible with the MPO-style E-step derivation. Following MPO [2], the E-step starts from the reference $q = \hat{\pi}_k$ and performs a one-step lookahead, so the Q-function used inside the E-step is $Q^{\hat{\pi}_k}$, in which the entropy bonus for $\hat{\pi}_k$ is already absorbed. The resulting energy

$$f_{\hat{s}, k}(a) = Q^{\hat{\pi}_k}(\hat{s}, a) - \alpha \log \tilde{\pi}_{\theta_k}(a | s) = Q^{\pi_k}(s, a) - \alpha \log \tilde{\pi}_{\theta_k}(a | s)$$

is a function of (s, a) alone, with θ_k treated as a constant within iteration k . Hence the MPO closed-form E-step solution and the subsequent M-step derivation apply without modification. The effect of the parameter update $\theta_k \rightarrow \theta_{k+1}$ between iterations (the reward drift) is analyzed separately in the monotonic-improvement proof (Appendix C.3).

510 To derive a tractable objective, we construct a variational lower bound on the log-likelihood of
511 optimality $\log p_{\pi}(\mathcal{O} = 1)$, following the RL-as-inference framework [24]. With the policy induced
512 trajectory $\tau \sim p_{\pi}(\tau)$ in Eq. (34), we apply Jensen’s inequality with an auxiliary distribution $p_q(\tau)$:
513

$$\begin{aligned} \log p_{\pi}(\mathcal{O} = 1) &= \log \int p_{\pi}(\tau) p(\mathcal{O} = 1 | \tau) d\tau \geq \int p_q(\tau) \left[\log p(\mathcal{O} = 1 | \tau) + \log \frac{p_{\pi}(\tau)}{p_q(\tau)} \right] d\tau \\ &= \mathbb{E}_q \left[\sum_t \gamma^t r_q(s_t, a_t) / \lambda \right] - D_{\text{KL}}(p_q(\tau) \| p_{\pi}(\tau)), \end{aligned}$$

514 Following the standard derivation in [2, 24], we absorb the entropy component of the trajectory-level
 515 KL into the per-step augmented reward

$$r_q(s_t, a_t) = r(s_t, a_t) + \alpha \mathbb{E}_{s_{t+1}} [\mathcal{H}(q(\cdot | s_{t+1}), \tilde{\pi}(\cdot | s_{t+1}))],$$

516 and express the remaining per-step KL in discounted form, yielding the variational objective

$$J(q, \xi) = \mathbb{E}_q \left[\sum_{t=0}^{\infty} \gamma^t \left(\frac{r_q(s_t, a_t)}{\lambda} - D_{\text{KL}}(q(\cdot | s_t) \| \pi(\cdot | s_t; \xi)) \right) \right].$$

517 We now extend the same variational argument to the z -MDP $\hat{\mathcal{M}}$, on which FLAG actually operates.
 518 Replacing the trajectory τ with the augmented trajectory $\hat{\tau}$ and using the marginal consistency
 519 established in Corollary 4.2, we obtain:

$$\begin{aligned} \log p_{\hat{\pi}}(\mathcal{O} = 1) &= \log \int p_{\hat{\pi}}(\hat{\tau}) p(\mathcal{O} = 1 | \hat{\tau}) d\hat{\tau} \geq \int p_q(\hat{\tau}) \left[\log p(\mathcal{O} = 1 | \hat{\tau}) + \log \frac{p_{\hat{\pi}}(\hat{\tau})}{p_q(\hat{\tau})} \right] d\hat{\tau} \\ &= \mathbb{E}_q \left[\sum_t \hat{r}_q(\hat{s}_t, a_t) / \lambda \right] - D_{\text{KL}}(p_q(\hat{\tau}) \| p_{\hat{\pi}}(\hat{\tau})) \\ \therefore J(q, \xi) &= \mathbb{E}_q \left[\sum_{t=0}^{\infty} \gamma^t [\hat{r}_q(\hat{s}_t, a_t) - \lambda D_{\text{KL}}(q(\cdot | \hat{s}_t) \| \hat{\pi}(\cdot | \hat{s}_t, \xi))] \right]. \end{aligned}$$

520 **Constrained E-step** We follow the MPO [2] derivation and adapt it to hard KL constraints and
 521 one-step bootstrapping. At iteration k , the objective function is

$$\max_q \mathbb{E}_{\hat{s} \sim \hat{\rho}_{\hat{\pi}}} [\mathbb{E}_{a \sim q(\cdot | \hat{s})} [Q^{\pi}(s, a) - \alpha \log \tilde{\pi}(a | s)]] \quad s.t. [D_{\text{KL}}(q(\cdot | \hat{s}) \| \hat{\pi}(\cdot | \hat{s}; \xi_k)) \leq \epsilon. \quad (41)$$

522 This objective has closed-form solution q_k , which is called the non-parametric variational distribution:

$$q_k(a | \hat{s}) = \frac{\hat{\pi}(a | \hat{s}; \xi_k) \exp(f_{\hat{s}, k}(a) / \lambda^*)}{Z(\hat{s})}, \quad Z(\hat{s}) = \int \hat{\pi}(a | \hat{s}; \xi_k) \exp(f_{\hat{s}, k}(a) / \lambda^*) da. \quad (42)$$

523 **M-step** In the M-step, given the fixed non-parametric target distribution q_k , we update the policy
 524 parameters ξ_k by projecting q_k back to our parametric rollout policy class:

$$\xi_{k+1} \in \arg \min_{\xi} \mathbb{E}_{\hat{\rho}_{\hat{\pi}_k}} [D_{\text{KL}}(q_k(\cdot | \hat{s}) \| \hat{\pi}(\cdot | \hat{s}; \xi_k))]. \quad (43)$$

525 C Proofs

526 **Standing Assumptions.** We consider an infinite-horizon γ -discounted MDP $\mathcal{M} =$
 527 $(\mathcal{S}, \mathcal{A}, p, r, \rho_0, \gamma)$. Fix a reference measure μ on $\mathcal{A}$ and assume all policies admit densities with
 528 respect to μ . Throughout this appendix, we write da in place of $d\mu(a)$ for notational simplicity.
 529 Accordingly, $\tilde{\pi}(\cdot | s)$ and $\pi(\cdot | s)$ denote μ -densities, and all KL divergences and expectations are
 530 taken with respect to these densities. Furthermore, we assume that $\tilde{\pi}$ and π are differentiable in $\mathcal{A}$.

Assumption C.1 (Bounded Augmented Reward). The entropy-augmented reward is uniformly bounded. Specifically, there exists a constant $R_{\max}^{\text{aug}} < \infty$ such that

$$|r(s, a) + \alpha b(s, a)| \leq R_{\max}^{\text{aug}}, \quad \forall (s, a), \quad (44)$$

where $b(s, a)$ denotes the entropy or cross-entropy bonus term (e.g., $-\log \tilde{\pi}(a | s)$).

Assumption C.2 (Regularity of Base Density). For each state s , the base density $\tilde{\pi}(\cdot | s)$ is three-times continuously differentiable with respect to a and is strictly positive μ -a.e. Moreover, the magnitudes of its third-order partial derivatives are pointwise bounded by a nonnegative

function $M_s(a)$ satisfying the integrability condition:

$$\int_{\mathcal{A}} M_s(a) da < \infty. \quad (45)$$

Assumption C.3 (Integrable gradient of the base density). For each state s , the gradient of the base density $\tilde{\pi}(\cdot | s)$ w.r.t a is integrable:

$$\|\nabla_a \tilde{\pi}(\cdot | s)\|_{L^1} \triangleq \int_{\mathcal{A}} \|\nabla_a \tilde{\pi}(a | s)\|_2 da < \infty. \quad (46)$$

This condition ensures that the density $\tilde{\pi}$ does not exhibit unbounded variation in the action space.

Assumption C.4 (Finite Partition Function). For every state $\hat{s}$ and iteration k , the function $Q^k(\hat{s}, \cdot)$ is measurable, and the corresponding Boltzmann distribution is well-defined (i.e., normalizable):

$$\int_{\mathcal{A}} \exp\left(\frac{Q^k(\hat{s}, a)}{\alpha}\right) da < \infty. \quad (47)$$

C.1 Proof of Proposition 4.3

In this section assume that the Gaussian smoothing covariance is state-independent and isotropic. For notational simplicity, we denote it by Σ , with $\Sigma = \sigma^2 I$ and denote $\tilde{\pi}_\theta$ simply as $\tilde{\pi}$ hereafter. We prove Proposition 4.3 in two parts: Lemma C.5 establishes the TV bound, and Lemma C.6 establishes the second-order KL bound.

Let $\delta \sim \mathcal{N}(0, \Sigma)$ be an independent noise at $a = T_\theta(s, z) + \delta$. Mathematically, this takes the form

$$\pi(\cdot | s) = \tilde{\pi}(\cdot | s) * \varphi_\Sigma, \quad (48)$$

where $*$ is the convolution sign in the action space.

C.1.1 Total variation distance bound

We assume that the following inequality holds for all s :

$$\|\nabla_a \tilde{\pi}(\cdot | s)\|_{L^1} \triangleq \int \|\nabla_a \tilde{\pi}(a | s)\|_2 da < \infty. \quad (49)$$

Lemma C.5. Suppose the global policy admits a Gaussian-kernel convolution form

$$\pi(a | s) = \int \tilde{\pi}(a - \delta | s) \varphi_\Sigma(\delta) d\delta, \quad (50)$$

where φ_Σ is the density of $\mathcal{N}(0, \Sigma)$. Then the following inequality holds for all $s \in \mathcal{S}$

$$D_{\text{TV}}(\pi(\cdot | s), \tilde{\pi}(\cdot | s)) \leq \frac{1}{2} \|\nabla_a \tilde{\pi}(\cdot | s)\|_{L^1} \mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma)} [\|\delta\|_2]. \quad (51)$$

Proof. By definition of the convolution in Eq. (50),

$$\pi(a | s) - \tilde{\pi}(a | s) = \int (\tilde{\pi}(a - \delta | s) - \tilde{\pi}(a | s)) \varphi_\Sigma(\delta) d\delta. \quad (52)$$

Using the triangle inequality,

$$\|\pi(\cdot | s) - \tilde{\pi}(\cdot | s)\|_{L^1} = \int \left| \int (\tilde{\pi}(a - \delta | s) - \tilde{\pi}(a | s)) \varphi_\Sigma(\delta) d\delta \right| da \quad (53)$$

$$\leq \int \int |\tilde{\pi}(a - \delta | s) - \tilde{\pi}(a | s)| \varphi_\Sigma(\delta) d\delta da \quad (54)$$

Since the integrand is nonnegative, Tonelli's theorem yields

$$\|\pi(\cdot | s) - \tilde{\pi}(\cdot | s)\|_{L^1} \leq \int \varphi_\Sigma(\delta) \left(\int |\tilde{\pi}(a - \delta | s) - \tilde{\pi}(a | s)| da \right) d\delta. \quad (55)$$

549 For each fixed (a, δ) , by the integral form of the mean value theorem,

$$\tilde{\pi}(a - \delta | s) - \tilde{\pi}(a | s) = - \int_0^1 \delta^\top \nabla_a \tilde{\pi}(a - t\delta | s) dt. \quad (56)$$

550 Applying the Cauchy–Schwarz inequality yields

$$|\tilde{\pi}(a - \delta | s) - \tilde{\pi}(a | s)| \leq \|\delta\|_2 \int_0^1 \|\nabla_a \tilde{\pi}(a - t\delta | s)\|_2 dt. \quad (57)$$

551 Integrating over a and using the change of variables $u = a - t\delta$,

$$\begin{aligned} \int |\tilde{\pi}(a - \delta | s) - \tilde{\pi}(a | s)| da &\leq \|\delta\|_2 \int_0^1 \int \|\nabla_a \tilde{\pi}(a - t\delta | s)\|_2 da dt \\ &= \|\delta\|_2 \int_0^1 \int \|\nabla_a \tilde{\pi}(u | s)\|_2 du dt \\ &= \|\delta\|_2 \|\nabla_a \tilde{\pi}(\cdot | s)\|_{L^1}. \end{aligned} \quad (58)$$

552 Plugging this into Eq. (55) bound gives

$$\|\pi(\cdot | s) - \tilde{\pi}(\cdot | s)\|_{L^1} \leq \|\nabla_a \tilde{\pi}(\cdot | s)\|_{L^1} \mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma)}[\|\delta\|_2].$$

553 Finally, using $D_{\text{TV}}(p, q) = \frac{1}{2} \|p - q\|_{L^1}$ concludes the proof. Moreover, by Cauchy–Schwarz,

$$\mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma)}[\|\delta\|_2] \leq \sqrt{\mathbb{E}[\|\delta\|_2^2]} = \sqrt{\mathbb{E}[\delta^\top \delta]} = \sqrt{\text{tr}(\Sigma)}.$$

554 Combining the above bounds implies $D_{\text{TV}}(\pi, \tilde{\pi}) = \mathcal{O}(\sqrt{\text{tr}(\Sigma)})$ as $\text{tr}(\Sigma) \rightarrow 0$. $\square$

555 C.1.2 Kullback-Leibler Divergence Bound

556 In this section, we show that when $\pi(\cdot | s)$ is obtained by smoothing $\tilde{\pi}(\cdot | s)$ with a *small-variance*
557 Gaussian kernel in Eq. (48), the discrepancy term $D_{\text{KL}}(\pi \| \tilde{\pi})$ vanishes at second order in $\text{tr}(\Sigma)$.

558 Recall the identity

$$\mathcal{H}(\pi, \tilde{\pi}) = \mathcal{H}(\pi) + D_{\text{KL}}(\pi \| \tilde{\pi}),$$

559 where $\mathcal{H}(\pi, \tilde{\pi}) \triangleq \mathbb{E}_\pi[-\log \tilde{\pi}]$ is the cross-entropy.

Lemma C.6 (Second-order KL bound under small Gaussian smoothing). *For each state s , let $\tilde{\pi}(\cdot | s)$ satisfy Assumption C.2 and strictly positive in a . Assume that the global policy admits the Gaussian smoothing form in Eq. (50). In addition, we assume the integrability condition*

$$\int \frac{(\text{tr}(\Sigma \nabla_a^2 \tilde{\pi}(a | s)))^2}{\tilde{\pi}(a | s)} da < \infty. \quad (59)$$

Then, as $\text{tr}(\Sigma) \rightarrow 0$,

$$D_{\text{KL}}(\pi(\cdot | s) \| \tilde{\pi}(\cdot | s)) = \mathcal{O}(\text{tr}(\Sigma)^2). \quad (60)$$

Consequently,

$$\mathcal{H}(\pi, \tilde{\pi}) = \mathcal{H}(\pi) + \mathcal{O}(\text{tr}(\Sigma)^2),$$

i.e., the cross-entropy approximates the true entropy up to a second-order error in $\text{tr}(\Sigma)$.

560 *Proof.* The smoothing form Eq. (50) can be written as an expectation over the Gaussian perturbation:

$$\pi(a | s) = \mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma)} [\tilde{\pi}(a - \delta | s)]. \quad (61)$$

562 When Σ is small, the random shift δ concentrates near 0. We therefore Taylor-expand $\tilde{\pi}(\cdot | s)$ around
563 a and evaluate at $a - \delta$:

$$\tilde{\pi}(a - \delta | s) = \tilde{\pi}(a | s) - \nabla_a \tilde{\pi}(a | s)^\top \delta + \frac{1}{2} \delta^\top \nabla_a^2 \tilde{\pi}(a | s) \delta + \mathcal{O}(\|\delta\|_2^3). \quad (62)$$

564 Taking expectation of Eq. (62) w.r.t. $\delta \sim \mathcal{N}(0, \Sigma)$ yields:

$$\pi(a | s) = \tilde{\pi}(a | s) - \mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma)} [\nabla_a \tilde{\pi}(a | s)^\top \delta] + \mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma)} \left[\frac{1}{2} \delta^\top \nabla_a^2 \tilde{\pi}(a | s) \delta \right] + \mathcal{O}(\|\delta\|_2^3).$$

565 The first-order term vanishes because $\mathbb{E}[\delta] = 0$, and the second-order term becomes a trace term
566 because $\mathbb{E}[\delta \delta^\top] = \Sigma$:

$$\mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma)} [\delta^\top \nabla_a^2 \tilde{\pi}(a | s) \delta] = \text{tr}(\Sigma \nabla_a^2 \tilde{\pi}(a | s)).$$

567 Therefore, a third-order Taylor expansion yields

$$\pi(a | s) = \tilde{\pi}(a | s) + \frac{1}{2} \text{tr}(\Sigma \nabla_a^2 \tilde{\pi}(a | s)) + R_3(a, s), \quad (63)$$

568 where the remainder satisfies $|R_3(a, s)| \leq C(s) \mathbb{E}[\|\delta\|_2^3]$. Applying the Cauchy–Schwarz inequality
569 and Isserlis’ theorem for Gaussian moments, we obtain

$$\mathbb{E}[\|\delta\|_2^3] \leq (\mathbb{E}[\|\delta\|_2^2])^{1/2} (\mathbb{E}[\|\delta\|_2^4])^{1/2} = \mathcal{O}(\text{tr}(\Sigma)^{1/2} \cdot \text{tr}(\Sigma)) = \mathcal{O}(\text{tr}(\Sigma)^{3/2}),$$

570 implying that the remainder term is of order $\mathcal{O}(\text{tr}(\Sigma)^{3/2})$. Note that the remainder’s contribution to
571 $(\pi - \tilde{\pi})$ at order $\mathcal{O}(\text{tr}(\Sigma)^{3/2})$ becomes $\mathcal{O}(\text{tr}(\Sigma)^3)$ after squaring.

572 To establish the order of the KL divergence, we analyze the relative perturbation of the rollout policy
573 π with respect to the base density $\tilde{\pi}$:

$$x(a) \triangleq \frac{\pi(a | s) - \tilde{\pi}(a | s)}{\tilde{\pi}(a | s)}. \quad (64)$$

574 By construction, $\pi(\cdot | s)$ is obtained by convolving $\tilde{\pi}(\cdot | s)$ with a small-variance Gaussian kernel. In
575 addition, we assume a uniform relative closeness condition: for sufficiently small Σ ,

$$\|x\|_\infty \leq c \text{tr}(\Sigma) \quad (65)$$

576 for some constant $c > 0$, which is chosen arbitrarily. Hence, $|x(a)| \leq 1/2$ for all $a \in \mathcal{A}$ whenever
577 $\text{tr}(\Sigma) \leq (2c)^{-1}$, justifying the uniform Taylor expansion

$$\log(1 + x) = x - \frac{x^2}{2} + R_3(x), \quad |R_3(x)| \leq C|x|^3 \text{ for } |x| \leq \frac{1}{2}. \quad (66)$$

578 Using $\pi = \tilde{\pi}(1 + x)$, we expand

$$\begin{aligned} D_{\text{KL}}(\pi \| \tilde{\pi}) &= \int \pi(a | s) \log \frac{\pi(a | s)}{\tilde{\pi}(a | s)} da \\ &= \int \tilde{\pi}(a | s) (1 + x(a)) \log(1 + x(a)) da \\ &= \int \tilde{\pi}(a | s) \left(x(a) + \frac{x(a)^2}{2} \right) da + \int \tilde{\pi}(a | s) (1 + x(a)) R_3(x(a)) da. \end{aligned} \quad (67)$$

579 The linear term vanishes since $\int \tilde{\pi} x da = \int (\pi - \tilde{\pi}) da = 0$. For the remainder term, $|x| \leq 1/2$
580 implies $1 + x \leq 3/2$ and therefore

$$\left| \int \tilde{\pi}(1 + x) R_3(x) da \right| \leq \frac{3C}{2} \int \tilde{\pi} |x|^3 da \leq \frac{3C}{2} \|x\|_\infty \int \tilde{\pi} x^2 da.$$

581 Consequently,

$$D_{\text{KL}}(\pi \| \tilde{\pi}) = \frac{1}{2} \int \tilde{\pi}(a | s) x(a)^2 da + \mathcal{O}\left(\|x\|_\infty \int \tilde{\pi} x^2 da\right). \quad (68)$$

582 Consequently, by Eq. (63) and Eq. (68), the leading term of $x(a)^2$ satisfies

$$x(a)^2 = \frac{1}{4} \left(\frac{\text{tr}(\Sigma \nabla_a^2 \tilde{\pi}(a | s))}{\tilde{\pi}(a | s)} \right)^2 + \text{higher-order terms}.$$

583 Under the integrability condition in Eq. (59), this yields

$$\int \tilde{\pi} x^2 da = \mathcal{O}(\text{tr}(\Sigma)^2).$$

584 Combining this with Eq. (65) and Eq. (68) proves $D_{\text{KL}}(\pi \| \tilde{\pi}) = \mathcal{O}(\text{tr}(\Sigma)^2)$. $\square$

C.2 Monotonic Improvement by Soft Actor Critic

In this section, we establish the SAC perspective of Theorem 4.5, which connects FLAG’s moment-matching update to SAC-style soft policy improvement. The argument proceeds in two steps. First, we show that under a sufficiently small local covariance Σ , the composite policy π remains close to the base flow policy $\tilde{\pi}$ in the sense of soft Q-function (Appendix C.2.1). This allows us to treat the SAC soft Bellman backup on $\tilde{\pi}$ as the reference dynamics for policy improvement. Second, we show that the moment-matching target $\mu_k^*(\hat{s})$ from Eq. (24) approximates the action gradient of the SAC objective $f_{\hat{s},k}$ via log-sum-exp trick, producing a zeroth-order BPTT-free estimate of the update direction that SAC realizes through the reparameterization-based action gradient (Appendix C.2.2).

C.2.1 Q-function Closeness between Composite Policy and Flow Policy

To show that the difference between the expected sum of returns with the policy π and those for the base flow policy $\tilde{\pi}$, first we need to look at how difference the discounted state distributions $\rho_\pi(s), \rho_{\tilde{\pi}}(s)$ are. This section refers to the proof of TRPO [39].

Lemma C.7. *The difference between the state distributions induced by π and $\tilde{\pi}$ has the following bound:*

$$\|\rho_\pi - \rho_{\tilde{\pi}}\|_1 \leq \frac{2\gamma\delta_{\text{TV}}}{(1-\gamma)^2}, \quad (69)$$

where δ_{TV} is defined as

$$\delta_{\text{TV}} \triangleq D_{\text{TV}}^{\max}(\pi, \tilde{\pi}) = \sup_s D_{\text{TV}}(\pi(\cdot | s), \tilde{\pi}(\cdot | s)) \leq \frac{1}{2} \left(\sup_s \|\nabla_a \tilde{\pi}(\cdot | s)\|_{L^1} \right) \cdot \sqrt{\text{tr}(\Sigma)},$$

by leveraging Lemma C.5

Proof. Let d_t^π denote the state distribution in timestep t , then we can bound the total variation distance between d_t^π and $d_t^{\tilde{\pi}}$ as follows

$$D_{\text{TV}}(d_t^\pi, d_t^{\tilde{\pi}}) \leq P(n_t > 0) \leq 1 - (1 - \delta_{\text{TV}})^t \leq t\delta_{\text{TV}}, \quad (70)$$

where n_t denote the number of times that $a_i \neq \tilde{a}_i$ for i , i.e. the number of times that π and $\tilde{\pi}$ disagree before timestep t . See [39], Lemma 3 for more details.

The total variation distance between the discounted visitation measure between π and $\tilde{\pi}$ is represented with the state distribution d_t :

$$\|\rho_\pi - \rho_{\tilde{\pi}}\|_1 = \left\| \sum_{t=0}^{\infty} \gamma^t (d_t^\pi - d_t^{\tilde{\pi}}) \right\|_1. \quad (71)$$

By triangle inequality,

$$\|\rho_\pi - \rho_{\tilde{\pi}}\|_1 \leq \sum_{t=0}^{\infty} \gamma^t \|d_t^\pi - d_t^{\tilde{\pi}}\|_1. \quad (72)$$

Since $\|p - q\|_1 = 2D_{\text{TV}}(p, q)$,

$$\|\rho_\pi - \rho_{\tilde{\pi}}\|_1 \leq \sum_{t=0}^{\infty} \gamma^t \|d_t^\pi - d_t^{\tilde{\pi}}\|_1 = 2 \sum_{t=0}^{\infty} \gamma^t D_{\text{TV}}(d_t^\pi, d_t^{\tilde{\pi}}) \quad (73)$$

$$= 2 \sum_{t=0}^{\infty} \gamma^t (t\delta_{\text{TV}}) = 2\gamma\delta_{\text{TV}} \sum_{t=1}^{\infty} t\gamma^{t-1} = \frac{2\gamma\delta_{\text{TV}}}{(1-\gamma)^2}. \quad (74)$$

607

□

This inequality states that if δ_{TV} is bounded $\delta_{\text{TV}} = \mathcal{O}(\sqrt{\text{tr}(\Sigma)})$, then the difference of the discounted measure between π and $\tilde{\pi}$ is bounded with $\mathcal{O}(\sqrt{\text{tr}(\Sigma)})$.

Next, using Lemma C.7, we bound the Q-function discrepancy between the composite policy π and the base flow policy $\tilde{\pi}$ under their respective soft Bellman operators.

Lemma C.8 (Q-function closeness between π and $\tilde{\pi}$). *Let Q^π and $Q^{\tilde{\pi}}$ denote the soft Q-functions associated with the composite policy π (under the **cross-entropy-regularized soft Bellman operator** $\mathcal{T}^\pi$) and the base flow policy $\tilde{\pi}$ (under the **entropy-regularized soft Bellman operator** $\mathcal{T}^{\tilde{\pi}}$), respectively. Under the standing assumptions,*

$$\|Q^\pi - Q^{\tilde{\pi}}\|_\infty \leq \frac{2\delta_{\text{TV}}}{1-\gamma} \left(\alpha C_R + \frac{\gamma R_{\max}}{1-\gamma} \right) = \mathcal{O} \left(\frac{\sqrt{\text{tr}(\Sigma)}}{(1-\gamma)^2} \right), \quad (75)$$

where $C_R = \sup_{s,a} |\log \tilde{\pi}(a | s)|$ and δ_{TV} is the maximum TV distance defined in Lemma C.7.

Proof. With the cross-entropy augmented reward $r_\pi(s, a)$ (Eq. (40)), the soft Bellman operator $\mathcal{T}^\pi$ in Eq. (17),

$$(\mathcal{T}^\pi Q)(s, a) = r(s, a) + \mathbb{E}_{s' \sim p(\cdot | s, a)} [\alpha H(\pi(\cdot | s'), \tilde{\pi}(\cdot | s')) + \gamma \mathbb{E}_{a' \sim \pi(\cdot | s')} [Q(s', a')]],$$

is a γ -contraction in the infinite norm:

$$\|\mathcal{T}^\pi Q - \mathcal{T}^\pi Q'\|_\infty \leq \gamma \|Q - Q'\|_\infty.$$

Please note that $\mathcal{T}^{\tilde{\pi}}$ is the soft Bellman operator in MaxEnt-RL, which uses true entropy, while $\mathcal{T}^\pi$ is the Soft Bellman operator using cross-entropy, i.e.,

$$(\mathcal{T}^{\tilde{\pi}} Q)(s, a) = r(s, a) + \mathbb{E}_{s' \sim p(\cdot | s, a)} [\alpha \mathcal{H}(\tilde{\pi}(\cdot | s')) + \gamma \mathbb{E}_{a' \sim \tilde{\pi}(\cdot | s')} [Q(s', a')]],$$

where the reward is augmented by the true entropy

$$r_{\tilde{\pi}}(s, a) = r(s, a) + \alpha \mathbb{E}_{a' \sim \tilde{\pi}(\cdot | s')} [\mathcal{H}(\tilde{\pi}(\cdot | s'))].$$

While the reward definitions differ, their discrepancy is controlled by the TV distance. Specifically, the difference is:

$$\begin{aligned} |r_\pi(s, a) - r_{\tilde{\pi}}(s, a)| &= \alpha |\mathbb{E}_{a' \sim \pi} [-\log \tilde{\pi}(a' | s)] - \mathbb{E}_{a' \sim \tilde{\pi}} [-\log \tilde{\pi}(a' | s)]| \\ &\leq 2\alpha \|\log \tilde{\pi}(\cdot | s)\|_\infty \cdot D_{\text{TV}}(\pi(\cdot | s), \tilde{\pi}(\cdot | s)) \\ &\leq 2\alpha C_R \delta_{\text{TV}}, \end{aligned} \quad (76)$$

where $C_R = \sup_{s,a} |\log \tilde{\pi}(a | s)|$ is a constant bounded by Assumption C.1.

Now, we decompose the difference between the optimal Q-functions:

$$\begin{aligned} \|Q^\pi - Q^{\tilde{\pi}}\|_\infty &= \|\mathcal{T}^\pi Q^\pi - \mathcal{T}^{\tilde{\pi}} Q^{\tilde{\pi}}\|_\infty \\ &\leq \|\mathcal{T}^\pi Q^\pi - \mathcal{T}^\pi Q^{\tilde{\pi}}\|_\infty + \|\mathcal{T}^\pi Q^{\tilde{\pi}} - \mathcal{T}^{\tilde{\pi}} Q^{\tilde{\pi}}\|_\infty. \end{aligned} \quad (77)$$

The first term is bounded by the contraction property of the Bellman operator:

$$\|\mathcal{T}^\pi Q^\pi - \mathcal{T}^\pi Q^{\tilde{\pi}}\|_\infty \leq \gamma \|Q^\pi - Q^{\tilde{\pi}}\|_\infty.$$

For the second term, we analyze the operator difference at any state-action pair (s, a) :

$$|(\mathcal{T}^\pi Q^{\tilde{\pi}} - \mathcal{T}^{\tilde{\pi}} Q^{\tilde{\pi}})(s, a)| \leq |r_\pi(s, a) - r_{\tilde{\pi}}(s, a)| + \gamma |\mathbb{E}_{s'} [\mathbb{E}_{a' \sim \pi} [Q^{\tilde{\pi}}(s', a')] - \mathbb{E}_{a' \sim \tilde{\pi}} [Q^{\tilde{\pi}}(s', a')]]|.$$

Using the reward bound derived above and the definition of total variation distance for the expectation term:

$$\begin{aligned} \|\mathcal{T}^\pi Q^{\tilde{\pi}} - \mathcal{T}^{\tilde{\pi}} Q^{\tilde{\pi}}\|_\infty &\leq 2\alpha C_R \delta_{\text{TV}} + \gamma \cdot 2 \|Q^{\tilde{\pi}}\|_\infty \delta_{\text{TV}} \\ &= 2\delta_{\text{TV}} (\alpha C_R + \gamma \|Q^{\tilde{\pi}}\|_\infty). \end{aligned} \quad (78)$$

Substituting these back into the triangle inequality (Eq. (77)) yields:

$$(1 - \gamma) \|Q^\pi - Q^{\tilde{\pi}}\|_\infty \leq 2\delta_{\text{TV}} (\alpha C_R + \gamma \|Q^{\tilde{\pi}}\|_\infty). \quad (79)$$

Since $\|Q^{\tilde{\pi}}\|_\infty \leq \frac{R_{\max}}{1-\gamma}$ and $\delta_{\text{TV}} = \mathcal{O}(\sqrt{\text{tr}(\Sigma)})$ completes the proof. $\square$

C.2.2 SNIS Approach and Zeroth-Order Gradient

Fix an augmented state $\hat{s} = (s, z)$ and let the current local policy be Gaussian:

$$a = \mu + \delta, \quad \mu = T_{\theta_k}(s, z), \quad \delta \sim \mathcal{N}(0, \Sigma_k).$$

In the E-step of iteration k , the non-parametric target q_k takes the form

$$q_k(a | \hat{s}) \propto \hat{\pi}(a | \hat{s}; \theta_k) \cdot \exp(f_{\hat{s},k}(a)/\lambda), \quad f_{\hat{s},k}(a) = Q^{\pi_k}(s, a) - \log \tilde{\pi}_{\theta_k}(a | s),$$

632 **Moment matching via SNIS.** The target mean in iteration k is given in Eq. (24). Using the
 633 reparameterization $a = \mu + \delta$, define $w(\delta) := \exp(f_{\hat{s},k}(\mu + \delta)/\lambda)$. Then

$$\mu^* = \frac{\mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [w(\delta) (\mu_k + \delta)]}{\mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [w(\delta)]}. \quad (80)$$

634 Hence the mean shift admits the normalized weighted-residual form

$$\Delta(\mu) := \mu^* - \mu = \frac{\mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [w(\delta) \delta]}{\mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [w(\delta)]}. \quad (81)$$

635 **Connection to a zeroth-order gradient.** In iteration k , we define the log-partition function $g(\mu)$:

$$g(\mu) := \log \mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [\exp(f(\mu + \delta)/\lambda)] = \log \mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [w(\delta)] \quad (82)$$

636 Differentiating g with respect to μ gives

$$\nabla_{\mu} g(\mu) = \frac{\nabla_{\mu} \mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [w(\delta)]}{\mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [w(\delta)]} = \frac{\mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [\nabla_{\mu} w(\delta)]}{\mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [w(\delta)]} \quad (83)$$

637 Since $a = \mu + \delta$ implies

$$\frac{\partial a}{\partial \delta} = I, \quad \frac{\partial a}{\partial \mu} = I \quad \longrightarrow \quad \nabla_{\delta} w(\delta) = \nabla_{\mu} w(\delta).$$

638 When we assume that $w(\delta)$ is sufficiently smooth and integrable with the boundary term equal to
 639 zero, we apply Stein's identity for $\delta \sim \mathcal{N}(0, \Sigma_k)$

$$\mathbb{E}_{\delta} [\delta w(\delta)] = \Sigma_k \mathbb{E}_{\delta} [\nabla_{\delta} w(\delta)]. \quad (84)$$

640 Combining Eq. (83) and Eq. (84) yields

$$\nabla_{\mu} g(\mu) = \frac{\mathbb{E}_{\delta} [\nabla_{\delta} w(\delta)]}{\mathbb{E}_{\delta} [w(\delta)]} = \Sigma_k^{-1} \frac{\mathbb{E}_{\delta} [\delta w(\delta)]}{\mathbb{E}_{\delta} [w(\delta)]} = \Sigma_k^{-1} \Delta(\mu). \quad (85)$$

641 Therefore, the direction to the target mean is the log-sum-exp estimate of zeroth-order gradient of
 642 $f_{\hat{s},k}(a)$:

$$\begin{aligned} \Delta(\mu) &= \Sigma_k \nabla_{\mu} g(\mu) \\ &= \Sigma_k \nabla_{\mu} \log \mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [\exp(f_{\hat{s},k}(\mu + \delta)/\lambda)]. \end{aligned} \quad (86)$$

Remark. From SAC's soft policy improvement to FLAG. The soft policy improvement theorem of SAC [20] states that any policy π_{new} minimizing

$$D_{\text{KL}} \left(\pi(a | s) \left\| \frac{\exp(Q^{\pi_{\text{old}}}/\alpha)}{Z} \right\| \right)$$

satisfies $Q^{\pi_{\text{new}}} \geq Q^{\pi_{\text{old}}}$. Importantly, the theorem is agnostic to how this KL minimizer is obtained. SAC restricts the policy class Π to a parametric Gaussian family and optimizes its parameters by gradient descent via reparameterization. In contrast, FLAG inherits the same improvement principle but solves the KL minimization through a non-parametric, action-level update (Figure 5).

Connection to the soft policy improvement theorem. Two ingredients align FLAG's update with the soft policy improvement theorem applied to the reference policy $\pi_{\text{old}} = \tilde{\pi}_{\theta_k}$.

- **Approximation of the soft objective.** Lemma C.8 shows that Q^{π} and $Q^{\tilde{\pi}_{\theta_k}}$ differ only by $\mathcal{O}(\sqrt{\text{tr}(\Sigma)})$. Therefore, FLAG's energy $f_{\hat{s},k}$ effectively realizes SAC's $\tilde{\pi}$ -based soft objective.
- **Approximation of the action-gradient update.** Appendix C.2.2 shows that the moment-matching direction

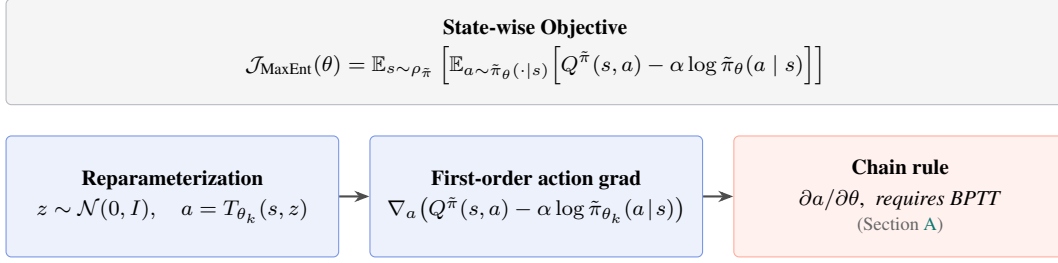
$$\mu_k^*(\hat{s}) - \mu_k = \Sigma_k \nabla_{\mu} g_k$$

is a zeroth-order approximation of SAC's first-order action gradient

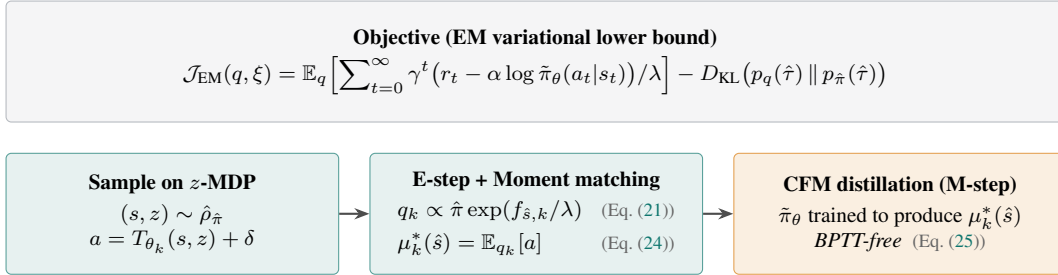
$$\nabla_a (Q^{\tilde{\pi}}(s, a) - \alpha \log \tilde{\pi}_{\theta_k}(a | s)),$$

via Stein's identity (Eq. (84)) and log-sum-exp smoothing (Eq. (86)).

SAC: parametric route via reparameterization



FLAG: non-parametric route via EM algorithm



How FLAG inherits SAC's soft policy improvement

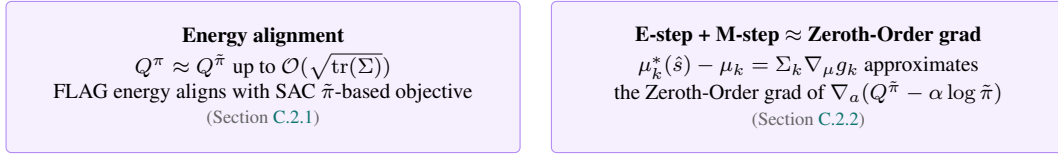


Figure 5: Comparison of SAC and FLAG at the k -th iteration. SAC realizes the soft policy improvement through reparameterization-based action gradients propagated to policy parameters via BPTT. FLAG instead solves an EM variational lower bound whose reward is regularized by cross-entropy. The two approaches are connected by (i) Q-function closeness between the composite policy π and base flow policy $\tilde{\pi}$ and (ii) the relation between FLAG's moment-matching update and a zeroth-order approximation of SAC's first-order action gradient.

Together, these two ingredients establish the SAC-based part of Theorem 4.5. The distinction between SAC's parametric route and FLAG's non-parametric route lies in the policy class Π and the corresponding optimization loss, not in the underlying soft policy improvement theorem.

What is distinctive about FLAG. The soft policy improvement theorem guarantees only that the target actions $\{\mu_k^*(\hat{s})\}_{\hat{s}}$ define an improved policy. It does not specify how a flow-based generative model should be trained to produce these actions. This is precisely where FLAG contributes: the M-step CFM distillation (Eq. (25)) provides a supervised, BPTT-free mechanism for training the flow policy to realize the improved target actions. In this way, FLAG provides a practical realization of SAC-style policy improvement for expressive flow-based policies.

C.3 Monotonic Improvement by Maximum a Posteriori Policy Optimisation

We now state a practically relevant monotonic-improvement result for FLAG in the latent-augmented MDP. We follow MPO [2] to prove the monotonic improvement in Eq. (19) and make some modifications since we are using cross-entropy augmented reward.

Setup and Notation. We work in a tabular z -MDP with finite augmented state space $\hat{\mathcal{S}}$ and a finite action grid $\mathcal{A}$. Each discrete action $a \in \mathcal{A}$ is identified with a bin $B_a \subset \mathbb{R}^{d_a}$ and a representative point $\bar{a} \in B_a$. The local Gaussian policy is defined on the underlying continuous action variable

652 $x \in \mathbb{R}^{d_a}$ and then projected onto the finite action grid:

$$\hat{\pi}_{\theta_k}(a \mid \hat{s}) := \int_{B_a} \varphi_{\Sigma_k}(x - \mu_{\theta_k}(\hat{s})) dx, \quad \Sigma_k = \sigma_k^2 I. \quad (87)$$

653 All Gaussian mean-shift and covariance calculations below are performed in this underlying continuous action space. we identify each discrete action $a \in \mathcal{A}$ with its representative point $\bar{a}$. In particular,
 654 $\mu_k^*(\hat{s}) := \sum_{a \in \mathcal{A}} q_k(a \mid \hat{s}) \bar{a}$. The induced discrete policy is obtained by bin projection. For notational
 655 simplicity, we write

$$\hat{\pi}_k(\cdot \mid \hat{s}) := \hat{\pi}_{\theta_k}(\cdot \mid \hat{s}), \quad \tilde{\pi}_k(\cdot \mid s) := \tilde{\pi}_{\theta_k}(\cdot \mid s).$$

657 The reward in iteration k is given as

$$\hat{r}_k(\hat{s}, a) := r(s, a) - \alpha \log \tilde{\pi}_k(a \mid s). \quad (88)$$

658 For a reference policy $\hat{\pi}$ and a non-parametric policy q , define the λ -regularized reward

$$\hat{r}_{k,\lambda}^{\hat{\pi}_k, q}(\hat{s}, a) := \hat{r}_k(\hat{s}, a) - \lambda \log \frac{q(a \mid \hat{s})}{\hat{\pi}_k(a \mid \hat{s})}. \quad (89)$$

659 The corresponding Bellman operators are

$$(\hat{\mathcal{T}}_k^q \hat{V})(\hat{s}) = \mathbb{E}_{a \sim q(\cdot \mid \hat{s})} \left[\hat{r}_k(\hat{s}, a) + \gamma \mathbb{E}_{\hat{s}' \sim \hat{p}(\cdot \mid \hat{s}, a)} [\hat{V}(\hat{s}')] \right], \quad (90)$$

$$(\hat{\mathcal{T}}_{k,\lambda}^{\hat{\pi}_k, q} \hat{V})(\hat{s}) = \mathbb{E}_{a \sim q(\cdot \mid \hat{s})} \left[\hat{r}_{k,\lambda}^{\hat{\pi}_k, q}(\hat{s}, a) + \gamma \mathbb{E}_{\hat{s}' \sim \hat{p}(\cdot \mid \hat{s}, a)} [\hat{V}(\hat{s}')] \right]. \quad (91)$$

660 We also define the corresponding value functions

$$\hat{V}_k^q(\hat{s}) = \mathbb{E}_q \left[\sum_{t=0}^{\infty} \gamma^t \hat{r}_k(\hat{s}_t, a_t) \mid \hat{s}_0 = \hat{s} \right], \quad (92)$$

$$\hat{V}_{k,\lambda}^{\hat{\pi}_k, q}(\hat{s}) = \mathbb{E}_q \left[\sum_{t=0}^{\infty} \gamma^t \left(\hat{r}_k(\hat{s}_t, a_t) - \lambda \log \frac{q(a_t \mid \hat{s}_t)}{\hat{\pi}_k(a_t \mid \hat{s}_t)} \right) \mid \hat{s}_0 = \hat{s} \right]. \quad (93)$$

661 Using the energy $f_{\hat{s},k}(a)$ defined in Eq. (20), the target of constrained E-step in Eq. (41) is given by

$$q_k(\cdot \mid \hat{s}) = \arg \max_q \mathbb{E}_{a \sim q(\cdot \mid \hat{s})} [f_{\hat{s},k}(a)] - \lambda (D_{\text{KL}}(q(\cdot \mid \hat{s}) \parallel \hat{\pi}_k(\cdot \mid \hat{s})) - \epsilon), \quad (94)$$

662 which admits the closed form

$$q_k(a \mid \hat{s}) = \frac{\hat{\pi}_k(a \mid \hat{s}) \exp(f_{\hat{s},k}(a)/\lambda)}{Z_k(\hat{s})}, \quad Z_k(\hat{s}) = \sum_{a \in \mathcal{A}} \hat{\pi}_k(a \mid \hat{s}) \exp(f_{\hat{s},k}(a)/\lambda). \quad (95)$$

663 Finally, define the exact KL-projection objective

$$h(\hat{\pi}, q, \theta) := \mathbb{E}_{\hat{\pi}, \hat{p}} \left[\sum_{t=0}^{\infty} \gamma^t D_{\text{KL}}(q(\cdot \mid \hat{s}_t) \parallel \hat{\pi}(\cdot \mid \hat{s}_t; \theta)) \mid \hat{s}_0, \hat{\pi} \right]. \quad (96)$$

Proposition C.9 (Bellman operator and Monotonicity). *For every iteration k , the E-step target q_k satisfies*

$$\hat{\mathcal{T}}_{k,\lambda}^{\hat{\pi}_k, q_k} \hat{V}_k^{\hat{\pi}_k} \geq \hat{V}_k^{\hat{\pi}_k}. \quad (97)$$

Consequently, by the monotonicity of the Bellman operator, the regularized value function satisfies the following improvement for all states $\hat{s}$:

$$\hat{V}_{k,\lambda}^{\hat{\pi}_k, q_k} \geq \hat{V}_k^{\hat{\pi}_k}. \quad (98)$$

664 *Proof.* The proof closely follows the policy evaluation argument presented in MPO, adapted to our
 665 latent-augmented z -MDP and cross-entropy reward structure. By the definition of q_k in Eq. (95), for
 666 every $\hat{s} \in \hat{\mathcal{S}}$, we have $(\hat{\mathcal{T}}_{k,\lambda}^{\hat{\pi}_k, q_k} \hat{V}_k^{\hat{\pi}_k})(\hat{s}) \geq (\hat{\mathcal{T}}_{k,\lambda}^{\hat{\pi}_k, \hat{\pi}_k} \hat{V}_k^{\hat{\pi}_k})(\hat{s}) = \hat{V}_k^{\hat{\pi}_k}$. Since the z -MDP is finite and
 667 the augmented reward $\hat{r}_k$ is bounded (Assumption C.1), the operator $\hat{\mathcal{T}}_{k,\lambda}^{\hat{\pi}_k, q_k}$ remains monotone and
 668 γ -contractive in the ∞ -norm. Therefore, repeatedly applying this operator and invoking the standard
 669 fixed-point theorem yields the final inequality $\hat{V}_{k,\lambda}^{\hat{\pi}_k, q_k} \geq \hat{V}_k^{\hat{\pi}_k}$. $\square$

671 **Objective decomposition.** Next, we establish an MPO-style one-step improvement bound for the
 672 EM update in Section 4.3. In standard MPO [2], the reward is fixed across the E-step and M-step,
 673 so the EM update can be interpreted as a coordinate-ascent step on a KL-regularized variational
 674 objective. In FLAG, the augmented reward

$$\hat{r}_k(\hat{s}, a) = r(s, a) - \alpha \log \tilde{\pi}_{\theta_k}(a | s)$$

675 also depends on the base flow policy. Therefore, after the base flow is updated from θ_k to θ_{k+1} , the
 676 reward changes from $\hat{r}_k$ to $\hat{r}_{k+1}$. The proof below separates these two effects:

$$\left(\underbrace{\text{frozen-reward MPO improvement}}_{\textcircled{1}} \right) + \left(\underbrace{\text{cross-entropy reward drift}}_{\textcircled{2}} \right). \quad (99)$$

677 For brevity, define

$$\mathcal{J}_k := \mathcal{J}(q_k, \theta_k) = \mathbb{E}_{\hat{s}_0 \sim \hat{p}} \left[\hat{V}_{k, \lambda}^{\hat{\pi}_k, q_k}(\hat{s}_0) \right] \quad (100)$$

678 where q_k denotes the exact non-parametric E-step target at iteration k . For theoretical analysis, we
 679 first consider the ideal KL-projection update

$$\theta_{k+1}^{\text{KL}} = \theta_k - \beta \nabla_{\theta} h(\hat{\pi}_k, q_k, \theta_k), \quad (101)$$

680 and denote the corresponding policy by $\hat{\pi}_{k+1}^{\text{KL}}(\cdot | \hat{s}) := \hat{\pi}(\cdot | \hat{s}; \theta_{k+1}^{\text{KL}})$. The practical CFM projection
 681 update is handled later in Lemma C.14.

Assumption C.10 (A Standard MPO regularity [2]). For each iteration k , the following hold:

1. The KL objective $h(\hat{\pi}_k, q_k, \theta_k)$ is L -smooth in a neighborhood of θ_k .
2. There exists $L_{\tilde{\pi}} > 0$ such that $\|\nabla_{\theta} \log \tilde{\pi}_{\theta}(a | s)\| \leq L_{\tilde{\pi}}$ for all θ in a neighborhood of θ_k .
3. The E-step target and the KL-projected policy change only to first order in the M-step size:

$$\sup_{\hat{s}} \|q_{k+1}(\cdot | \hat{s}) - q_k(\cdot | \hat{s})\|_1 = \mathcal{O}(\beta), \quad \sup_{\hat{s}} \|\hat{\pi}_{k+1}^{\text{KL}}(\cdot | \hat{s}) - \hat{\pi}_k(\cdot | \hat{s})\|_1 = \mathcal{O}(\beta).$$

Assumption C.11 (Variance-scaled drift alignment). Assume that, within iteration k , the local covariance is fixed with respect to θ , state-independent, and isotropic:

$$\Sigma_k = \sigma_k^2 I, \quad \sigma_k^2 > 0.$$

Here Σ_k denotes the covariance matrix for notational convenience, while σ_k^2 denotes its scalar variance. Let the first-order reward-drift score be

$$\mathcal{G}_k^{\tilde{\pi}} := \mathbb{E}_{q_k} \left[\sum_{t=0}^{\infty} \gamma^t \nabla_{\theta} \log \tilde{\pi}_{\theta_k}(a_t | s_t) \middle| \hat{s}_0 \right].$$

There exists a structural constant $C_{\Sigma} > 0$, independent of the step size β , such that

$$\left| \langle \mathcal{G}_k^{\tilde{\pi}}, \nabla_{\theta} h(\hat{\pi}_k, q_k, \theta_k) \rangle \right| \leq C_{\Sigma} \sigma_k^2 \mathcal{G}_k. \quad (102)$$

683 Given Assumption C.10 then for any $0 \leq \beta \leq 1/L$, the descent lemma [33] gives
 684

$$h(\hat{\pi}_k, q_k, \theta_{k+1}^{\text{KL}}) \leq h(\hat{\pi}_k, q_k, \theta_k) - \beta \mathcal{G}_k, \quad \mathcal{G}_k := \frac{1}{2} \|\nabla_{\theta} h(\hat{\pi}_k, q_k, \theta_k)\|^2. \quad (103)$$

685 This decrease in the KL projection objective will produce the standard MPO-style improvement when
 686 the reward is fixed at $\hat{r}_k$. The additional difficulty is that the cross-entropy term $-\alpha \log \tilde{\pi}_{\theta}(a | s)$ also
 687 changes after the update. The following two assumptions isolate these two components.

688 **Justification on Assumption C.11** While the following derivation relies on approximations, it
 689 establishes a structural bound on the drift term based on the zeroth-order approximation from
 690 Appendix C.2.2. This arises because the M-step parameter update $\Delta\theta_k$, which governs the drift,
 691 is shaped by the zeroth-order moment-matching direction. Following Section 4.1, we assume an
 692 isotropic covariance $\Sigma_k = \sigma_k^2 I$.

Let $\kappa_k(\hat{s}; \theta) = D_{\text{KL}}(q_k(\cdot | \hat{s}) \parallel \hat{\pi}(\cdot | \hat{s}; \theta))$ denote the statewise KL projection in the M-step. Using the zeroth-order approximation $\mu_k^*(\hat{s}) - \mu_k(\hat{s}) \approx \sigma_k^2 \nabla_{\mu} g_k(\mu_k(\hat{s}))$ from Eq. (86), the statewise M-step gradient with respect to θ is simplified to:

$$-\nabla_{\theta} \kappa_k(\hat{s}; \theta_k) = J_{\theta}(\hat{s})^{\top} \sigma_k^{-2} (\mu_k^*(\hat{s}) - \mu_k(\hat{s})) \approx J_{\theta}(\hat{s})^{\top} v_k(\hat{s}), \quad (104)$$

where $J_{\theta}(\hat{s}) := \partial \mu_k(\hat{s}) / \partial \theta$ and $v_k(\hat{s}) := \nabla_{\mu} g_k(\mu_k(\hat{s}))$. Consequently, the squared statewise gradient norm is independent of σ_k^{-2} :

$$\|\nabla_{\theta} \kappa_k(\hat{s}; \theta_k)\|^2 \approx v_k(\hat{s})^{\top} J_{\theta}(\hat{s}) J_{\theta}(\hat{s})^{\top} v_k(\hat{s}). \quad (105)$$

In contrast, applying a first-order Taylor expansion to the smoothed energy yields the statewise base-flow drift direction:

$$\begin{aligned} J_{\theta}(\hat{s})^{\top} \Delta \mu &= J_{\theta}(\hat{s})^{\top} \left(\mathbb{E}_{a \sim q_k(\cdot | \hat{s})} [a] - \mu_k(\hat{s}) \right) \\ &= J_{\theta}(\hat{s})^{\top} \frac{\mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [\delta \exp(f_{\hat{s}, k}(\mu_k + \delta)/\lambda)]}{\mathbb{E}_{\delta \sim \mathcal{N}(0, \Sigma_k)} [\exp(f_{\hat{s}, k}(\mu_k + \delta)/\lambda)]} \\ &= J_{\theta}(\hat{s})^{\top} \Sigma_k \nabla_{\mu} g_k(\mu_k(\hat{s})) \\ &= J_{\theta}(\hat{s})^{\top} \Sigma_k v_k(\hat{s}). \end{aligned} \quad (106)$$

To lift the statewise calculation to the global objective, we follow the standard MPO auxiliary-objective construction [2]. In the E-step, q_k is obtained as a statewise non-parametric improvement of the reference policy $\hat{\pi}_k$, while the subsequent projection is evaluated on states visited by $\hat{\pi}_k$. Therefore, we use the frozen discounted visitation measure $\hat{\rho}_k(\hat{s}) \equiv \hat{\rho}_{\hat{\pi}_k}(\hat{s})$ for the state weighting, and use $q_k(\cdot | \hat{s})$ only as the improved conditional action distribution at each visited state. Define the visitation-weighted mean-drift surrogate

$$\bar{G}_k^{\tilde{\pi}} := \sum_{\hat{s} \in \mathcal{S}} \hat{\rho}_k(\hat{s}) J_{\theta}(\hat{s})^{\top} \Delta \mu_k(\hat{s}). \quad (107)$$

This surrogate is the leading-order mean-shift approximation of the actual reward-drift score $\bar{G}_k^{\tilde{\pi}}$ in Assumption C.11.

Let $\mathbf{J}$ and $\mathbf{v}$ denote the corresponding state-stacked objects with weights $\sqrt{\hat{\rho}_k(\hat{s})}$. Then

$$U = \mathbf{J}^{\top} \mathbf{v}, \quad D \approx \sigma_k^2 \mathbf{J}^{\top} \mathbf{v}, \quad 2\mathcal{G}_k = \|U\|^2 = \mathbf{v}^{\top} \mathbf{J} \mathbf{J}^{\top} \mathbf{v}.$$

Hence the leading-order global alignment is

$$\langle \bar{G}_k^{\tilde{\pi}}, -\nabla_{\theta} h(\hat{\pi}_k, q_k, \theta_k) \rangle \approx D^{\top} U = \sigma_k^2 \mathbf{v}^{\top} (\mathbf{J} \mathbf{J}^{\top}) \mathbf{v} = 2\sigma_k^2 \mathcal{G}_k. \quad (108)$$

Thus the leading-order calculation gives the variance-scaled drift alignment. The factor 2, the mismatch between the actual reward-drift score $\bar{G}_k^{\tilde{\pi}}$ and the mean-drift surrogate $\bar{G}_k^{\tilde{\pi}}$, together with the zeroth-order, Taylor, and projection approximation errors, is absorbed into the constant C_{Σ} in Assumption C.11.

Proposition C.12 (Monotone Improvement in the z-MDP). *Under Assumptions C.10 and C.11, there exists a constant $C > 0$, independent of β , such that the ideal KL update satisfies*

$$\mathcal{J}_{k+1}^{\text{KL}} \geq \mathcal{J}_k + (\lambda - \alpha C_{\Sigma} \sigma_k^2) \beta \mathcal{G}_k - C \beta^2, \quad (109)$$

where

$$\mathcal{J}_{k+1}^{\text{KL}} := \mathbb{E}_{\hat{s}_0 \sim \hat{p}_0} \left[\hat{V}_{k+1, \lambda}^{\hat{\pi}_{k+1}^{\text{KL}}, q_{k+1}}(\hat{s}_0) \right].$$

In particular, if

$$\lambda > \alpha C_{\Sigma} \sigma_k^2, \quad (110)$$

then, for sufficiently small β ,

$$\mathcal{J}_{k+1}^{\text{KL}} \geq \mathcal{J}_k. \quad (111)$$

715 *Proof.*

716 **Step 1:** We first write the decomposition point-wise for a fixed initial state $\hat{s}_0$. Taking expectation
 717 over $\hat{s}_0 \sim \hat{p}_0$ then gives the corresponding statement for $\mathcal{J}_{k+1}^{\text{KL}} - \mathcal{J}_k$.

$$\begin{aligned} \mathcal{J}_{k+1}^{\text{KL}} - \mathcal{J}_k &= \mathbb{E}_{\hat{s}_0 \sim \hat{p}_0} \left[\hat{V}_{k+1, \lambda}^{\hat{\pi}_{k+1}^{\text{KL}}, q_{k+1}}(\hat{s}_0) - \hat{V}_{k, \lambda}^{\hat{\pi}_k, q_k}(\hat{s}_0) \right] \\ &= \mathbb{E}_{\hat{s}_0 \sim \hat{p}_0} \left[\underbrace{\left(\hat{V}_{k+1, \lambda}^{\hat{\pi}_{k+1}^{\text{KL}}, q_{k+1}}(\hat{s}_0) - \hat{V}_{k, \lambda}^{\hat{\pi}_k, q_k}(\hat{s}_0) \right)}_{\textcircled{1} \text{ in Eq. (99)}} + \underbrace{\left(\hat{V}_{k+1, \lambda}^{\hat{\pi}_{k+1}^{\text{KL}}, q_{k+1}}(\hat{s}_0) - \hat{V}_{k+1, \lambda}^{\hat{\pi}_{k+1}, q_{k+1}}(\hat{s}_0) \right)}_{\textcircled{2} \text{ in Eq. (99)}} \right]. \end{aligned} \quad (112)$$

718 The first bracket freezes the reward at $\hat{r}_k$, so it has exactly the same form as the objective analyzed in
 719 MPO. The second bracket is the only new term in FLAG, and it appears because the reward changes
 720 when $\tilde{\pi}_k$ is updated to $\tilde{\pi}_{k+1}$.

721 **Step 2:** Fix the iteration k and freeze the augmented reward $\hat{r}_k$. In this case, the z-MDP objective has
 722 the same KL-regularized coordinate-ascent structure as MPO, with the substitutions $s \mapsto \hat{s}$, $\pi \mapsto \hat{\pi}$,
 723 and $r \mapsto \hat{r}_k$. Accordingly, define the frozen-reward auxiliary functional

$$\mathcal{H}_k(\hat{\pi}, q, \theta, \hat{\pi}') := \mathbb{E}_{\hat{\pi}, \hat{p}} \left[\sum_{t=0}^{\infty} \gamma^t \left(\mathbb{E}_{a_t \sim q(\cdot | \hat{s}_t)} [\hat{Q}_k^{\hat{\pi}'}(\hat{s}_t, a_t)] - \lambda D_{\text{KL}}(q(\cdot | \hat{s}_t) \| \hat{\pi}(\cdot | \hat{s}_t; \theta)) \right) \right] \hat{s}_0. \quad (113)$$

724 where $Q_k^{\hat{\pi}'}(\hat{s}, a) := \hat{r}_k(\hat{s}, a) + \gamma \mathbb{E}_{\hat{s}' \sim \hat{p}(\cdot | \hat{s}, a)} [\hat{V}_k^{\hat{\pi}'}(\hat{s}')]$. By construction, $\mathcal{H}_k$ is exactly the MPO
 725 auxiliary objective specialized to the tabular z-MDP with frozen reward $\hat{r}_k$. Therefore, under As-
 726 sumption C.10, the argument of [2, Appendix A.2] applies to Eq. (113) and yields

$$\hat{V}_{k+1, \lambda}^{\hat{\pi}_{k+1}^{\text{KL}}, q_{k+1}}(\hat{s}_0) - \hat{V}_{k, \lambda}^{\hat{\pi}_k, q_k}(\hat{s}_0) \geq \lambda \beta \mathcal{G}_k - C_1 \beta^2, \quad (114)$$

727 for some constant $C_1 > 0$ independent of β . Equivalently, the frozen-reward part of the FLAG update
 728 inherits the same first-order improvement term $\lambda \beta \mathcal{G}_k$ as MPO, up to a second-order remainder $C_1 \beta^2$.

729 Strictly, the MPO argument exploits the one-step optimality of the E-step target with respect to
 730 the frozen reward $\hat{r}_k$, while the actual q_{k+1} in FLAG is optimal with respect to the iteration- $(k+1)$
 731 reward $\hat{r}_{k+1}$. The induced discrepancy is controlled by two first-order quantities: the Q-function drift
 732 and the E-step target drift. Since $\Delta \theta_k = \mathcal{O}(\beta)$, the reward drift $\hat{r}_{k+1} - \hat{r}_k$ is $\mathcal{O}(\beta)$, and combining
 733 this with Assumption C.10 gives

$$\|\hat{Q}_{k+1}^{\hat{\pi}_{k+1}} - \hat{Q}_k^{\hat{\pi}_{k+1}}\|_{\infty} = \mathcal{O}(\beta), \quad \sup_{\hat{s}} \|q_{k+1}(\cdot | \hat{s}) - q_k(\cdot | \hat{s})\|_1 = \mathcal{O}(\beta).$$

734 Transferring the optimality of q_{k+1} from the iteration- $(k+1)$ objective to the frozen-reward objective
 735 then introduces a single cross-term of these two quantities,

$$(\mathbb{E}_{q_{k+1}} - \mathbb{E}_{q_k}) [\hat{Q}_k^{\hat{\pi}_{k+1}} - \hat{Q}_{k+1}^{\hat{\pi}_{k+1}}] = \mathcal{O}(\beta^2),$$

736 which is absorbed into the constant C_1 in Eq. (114).

737 **Step 3:** Due to the moving base flow policy, the reward-drift term in Eq. (112) is given by

$$\hat{V}_{k+1, \lambda}^{\hat{\pi}_{k+1}^{\text{KL}}, q_{k+1}}(\hat{s}_0) - \hat{V}_{k, \lambda}^{\hat{\pi}_{k+1}^{\text{KL}}, q_{k+1}}(\hat{s}_0) = \alpha \mathbb{E}_{q_{k+1}} \left[\sum_{t=0}^{\infty} \gamma^t \left(\log \tilde{\pi}_k(a_t | s_t) - \log \tilde{\pi}_{k+1}^{\text{KL}}(a_t | s_t) \right) \right] \hat{s}_0. \quad (115)$$

738 We define this quantity as the reward-drift term Δ_k^{drift} .

739 Let the parameter update be $\Delta \theta_k := \theta_{k+1}^{\text{KL}} - \theta_k = -\beta \nabla_{\theta} h(\hat{\pi}_k, q_k, \theta_k)$. By Taylor expanding
 740 $\log \tilde{\pi}_{\theta_{k+1}^{\text{KL}}}(a | s)$ around θ_k , we obtain:

$$\begin{aligned} \Delta_k^{\text{drift}} &= -\alpha \mathbb{E}_{q_{k+1}} \left[\sum_{t=0}^{\infty} \gamma^t \nabla_{\theta} \log \tilde{\pi}_{\theta_k}(a_t | s_t) \right] \hat{s}_0^{\top} \Delta \theta_k + \mathcal{O}(\|\Delta \theta_k\|^2) \\ &= \alpha \beta \mathbb{E}_{q_{k+1}} \left[\sum_{t=0}^{\infty} \gamma^t \nabla_{\theta} \log \tilde{\pi}_{\theta_k}(a_t | s_t) \right] \hat{s}_0^{\top} \nabla_{\theta} h(\hat{\pi}_k, q_k, \theta_k) + \mathcal{O}(\beta^2). \end{aligned} \quad (116)$$

By Assumption C.10, we have $\sup_{\hat{s}} \|q_{k+1}(\cdot | \hat{s}) - q_k(\cdot | \hat{s})\|_1 = \mathcal{O}(\beta)$. By Assumption C.10 and the boundedness of the base-flow score, replacing the discounted score expectation under q_{k+1} by that under q_k changes the first-order Taylor coefficient by $\mathcal{O}(\beta)$. Since this coefficient is multiplied by β , the resulting error is $\mathcal{O}(\beta^2)$.

$$\Delta_k^{\text{drift}} = \alpha\beta \langle G_k^{\pi}, \nabla_{\theta} h(\hat{\pi}_k, q_k, \theta_k) \rangle + \mathcal{O}(\beta^2). \quad (117)$$

Plugging the Assumption C.11 into Eq. (117), the final bound on the reward-drift magnitude is

$$|\Delta_k^{\text{drift}}| \leq \alpha C_{\Sigma} \sigma_k^2 \beta \mathcal{G}_k + C_d \beta^2 \quad (118)$$

for some constant $C_d > 0$ independent of β .

Substituting the reward-freeze improvement (① in Eq. (114)) and reward-drift bound (② in Eq. (118)) into the objective decomposition (① + ② in Eq. (112)), we conclude:

$$\begin{aligned} \mathcal{J}_{k+1}^{\text{KL}} - \mathcal{J}_k &\geq \lambda \beta \mathcal{G}_k - C_1 \beta^2 - |\Delta_k^{\text{drift}}| \\ &\geq \lambda \beta \mathcal{G}_k - C_1 \beta^2 - (\alpha C_{\Sigma} \sigma_k^2 \beta \mathcal{G}_k + C_d \beta^2) \\ &= (\lambda - \alpha C_{\Sigma} \sigma_k^2) \beta \mathcal{G}_k - (C_1 + C_d) \beta^2. \end{aligned} \quad (119)$$

Hence, if

$$\lambda > \alpha C_{\Sigma} \sigma_k^2, \quad (120)$$

then, for a sufficiently small step size β , the overall objective is nondecreasing:

$$\mathcal{J}_{k+1}^{\text{KL}} \geq \mathcal{J}_k.$$

□

Practical CFM update. Proposition C.12 analyzes the KL-based M-step. In practice, however, FLAG distills the resulting target into the base flow policy via the CFM loss (Eq. (25)). To bound the gap, we relate the practical update to the KL update through an excess projection error, rather than claiming that the CFM loss itself decreases $h(\hat{\pi}_k, q_k, \theta_k)$.

Assumption C.13 (Approximate realization of the ideal KL update by CFM). There exists $\epsilon_k^{\text{proj}} \geq 0$ such that the practical CFM update satisfies

$$\mathcal{J}_{k+1}^{\text{CFM}} \geq \mathcal{J}_{k+1}^{\text{KL}} - \lambda \epsilon_k^{\text{proj}}, \quad (121)$$

where

$$\mathcal{J}_{k+1}^{\text{CFM}} := \mathbb{E}_{\hat{s}_0 \sim \hat{p}_0} \left[\hat{V}_{k+1, \lambda}^{\hat{\pi}_{k+1}^{\text{CFM}}, q_{k+1}}(\hat{s}_0) \right], \quad \mathcal{J}_{k+1}^{\text{KL}} := \mathbb{E}_{\hat{s}_0 \sim \hat{p}_0} \left[\hat{V}_{k+1, \lambda}^{\hat{\pi}_{k+1}^{\text{KL}}, q_{k+1}}(\hat{s}_0) \right].$$

Lemma C.14 (Approximate monotone improvement under the practical CFM update). Under Assumptions C.10, C.11 and C.13, there exists a constant $C > 0$, independent of β , such that

$$\mathcal{J}_{k+1}^{\text{CFM}} \geq \mathcal{J}_k + (\lambda - \alpha C_{\Sigma} \sigma_k^2) \beta \mathcal{G}_k - C \beta^2 - \lambda \epsilon_k^{\text{proj}}. \quad (122)$$

In particular, if

$$(\lambda - \alpha C_{\Sigma} \sigma_k^2) \beta \mathcal{G}_k \geq C \beta^2 + \lambda \epsilon_k^{\text{proj}}, \quad (123)$$

then

$$\mathcal{J}_{k+1}^{\text{CFM}} \geq \mathcal{J}_k.$$

Proof. By Assumption C.13,

$$\mathcal{J}_{k+1}^{\text{CFM}} \geq \mathcal{J}_{k+1}^{\text{KL}} - \lambda \epsilon_k^{\text{proj}}.$$

Applying Proposition C.12 to the ideal KL iterate yields

$$\mathcal{J}_{k+1}^{\text{KL}} \geq \mathcal{J}_k + (\lambda - \alpha C_{\Sigma} \sigma_k^2) \beta \mathcal{G}_k - C \beta^2.$$

Combining the two inequalities proves Eq. (122). The last statement follows immediately from Eq. (123). □

FLAG: MPO-style monotonic improvement

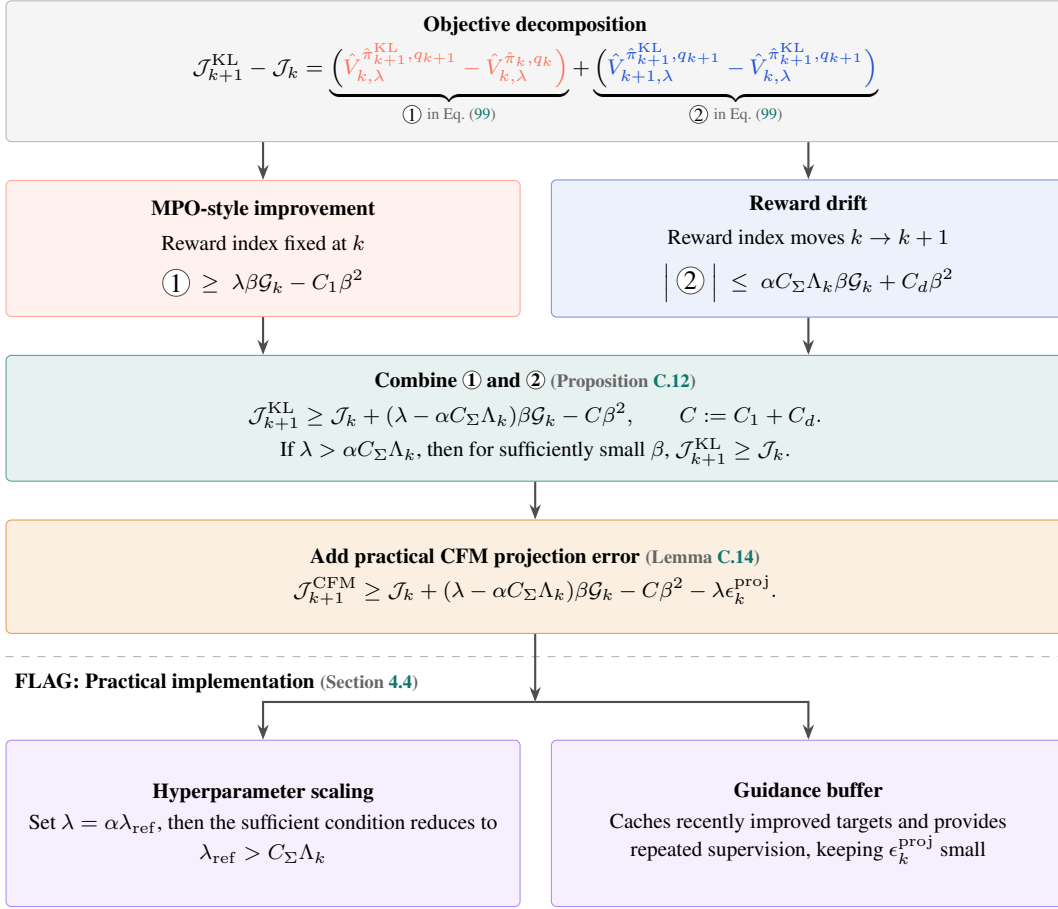


Figure 6: Proof roadmap for the MPO-style monotonic improvement of FLAG. The objective gap is decomposed into the standard MPO-style improvement term (① in Eq. (99)) and the FLAG-specific reward-drift term (② in Eq. (99)). Combining the two yields Proposition C.12, and adding the practical CFM projection error yields Lemma C.14. The bottom row shows two practical implementation choices that ensure the sufficient conditions: hyperparameter scaling $\lambda = \alpha \lambda_{\text{ref}}$ and the guidance buffer for keeping ϵ_k^{proj} small (Section 4.4).

Remark. Remark: Practical implications of the bound. The monotonic improvement guarantee in Proposition C.12 and Lemma C.14 carries two practical implications, both naturally satisfied by FLAG’s design (see Figure 6). First, using the effective temperature $\lambda = \alpha \lambda_{\text{ref}}$ (Section 4.4), the sufficient condition reduces to $\lambda_{\text{ref}} > C_\Sigma \sigma_k^2$, which holds because covariance scheduling or clipping keeps σ_k^2 small in practice. Second, the projection error ϵ_k^{proj} is kept modest by the guidance buffer, which caches recently improved targets and provides dense, repeated supervision for the flow policy across off-policy updates.

D Implementation Details

D.1 Action scaling and SNIS details

We map pre-action samples u to bounded actions $a \in [-1, 1]^{|A|}$ via an element-wise tanh, where $|A|$ is the cardinality of the action space. Let us denote the pre-action sampling distribution at iteration k as $r_k(u \mid \hat{s})$, which is a Gaussian centered on the flow policy output $\mu(\hat{s}; \theta_k)$ and has a scheduled covariance Σ_k . To apply change of variables to r_k , we need the determinant of the Jacobian (of the element-wise tanh), $|\det J_{\text{tanh}}(u)| = \prod_{i=1}^{|A|} (1 - a_i^2)$ where $a = \tanh(u)$. Applying change

of variables leads to the following:

$$\hat{\pi}_k(a \mid \hat{s}) = \frac{r_k(u \mid \hat{s})}{|\det J_{\tanh}(u)|}, \quad \log \hat{\pi}_k(a \mid \hat{s}) = \log r_k(u \mid \hat{s}) - \sum_{i=1}^D \log(1 - a_i^2). \quad (124)$$

We form the E-step target distribution following Eq. (21), only difference being the parameterization where we do not parameterize the covariance Σ_k , thus $\xi_k = \theta_k$. Pulling back to pre-action space via Eq. (124) yields

$$q_k(u \mid \hat{s}) = q_k(\tanh^{-1}(a) \mid \hat{s}) |\det J_{\tanh}(u)| \propto r_k(u \mid \hat{s}) \exp\left(\frac{f_{\hat{s},k}(\tanh(u))}{\alpha_k \lambda_{\text{ref}}}\right) \quad (125)$$

Using r_k as the proposal distribution, we define unnormalized importance weight w and subsequently self-normalized importance weight $\bar{w}$ with N pre-action samples. The moment matching in the pre-action space (u -space) is estimated by self-normalized importance sampling:

$$w(u) := \exp\left(\frac{f_{\hat{s},k}(\tanh(u))}{\alpha_k \lambda_{\text{ref}}}\right), \quad \bar{w}_i = \frac{w(u_i)}{\sum_{j=1}^N w(u_j)}, \quad \mu^*(\hat{s}) = \sum_{i=1}^N \bar{w}_i u_i. \quad (126)$$

D.2 Critic networks

We adapt the distributional Q-function from [5] to account for the entropy of the policy. The TD-target y in Eq. (17) is computed by

$$y = r(s, a) + \gamma \mathbb{E}_{a' \sim \pi(\cdot \mid s')} \left[\sum_{i=0}^{b-1} \left(Q_{\min} + \left(\frac{Q_{\max} - Q_{\min}}{b-1} \right) i \right) \cdot p_i - \alpha \log \tilde{\pi}_{\theta}(a' \mid s') \right], \quad (127)$$

where b is the number of bins and p_i denotes the probability of the i -th bins which is the output of the critic network. After calculating the target we project it to discrete bins, $\hat{y} = \text{two_hot}(y)$. The critic minimizes the following loss, which was proposed in [11].

$$\mathcal{L}_Q(\phi) = - \sum_{i=1}^b \hat{y} \log p_i - 0.005 \sum_{i=1}^b p_i \log p_i, \quad (128)$$

Following [31], we use the *mean* of the two TD-targets from two Q-function instead of *min*.

D.3 Variance Reduction in Hutchinson Trace Estimation

Rademacher Distribution. We sample ϵ from the Rademacher distribution, which typically yields a lower variance estimator [21, 3].

Common Random Numbers (CRN). When performing the estimation in Eq. 28, we evaluate the difference in log-probabilities between an action a and its perturbed neighbors $a + \delta$ to reduce the variance of the estimate compared to the case of estimating them individually. Specifically, the variance of the paired difference estimate can be minimized by employing the *Common Random Numbers* (CRN) technique [18]. This technique uses the exact same noise vector ϵ for both a and $a + \delta$, inducing a strong coupling between the two stochastic estimates.

Let $\hat{L}(a; \epsilon) = \epsilon^\top \nabla u(a) \epsilon$ denote the stochastic estimator for the trace term at a using a noise vector ϵ , where u is a vector field. The variance of the paired difference estimator is given by:

$$\text{Var}(\hat{L}(a + \delta; \epsilon) - \hat{L}(a; \epsilon)) = \text{Var}(\hat{L}(a + \delta; \epsilon)) + \text{Var}(\hat{L}(a; \epsilon)) - 2 \text{Cov}(\hat{L}(a + \delta; \epsilon), \hat{L}(a; \epsilon)). \quad (129)$$

Assuming that the Jacobian $\nabla u(a)$ varies smoothly when δ is small, $\hat{L}(a + \delta; \epsilon)$ and $\hat{L}(a; \epsilon)$ are positively correlated under the shared ϵ . Consequently, the large covariance term cancels a substantial portion of the individual variances, yielding a lower-variance estimate of the difference.

Application to Weight Calculation. To leverage the variance-reducing property of CRN, we formulate the importance sampling weights in terms of *differences*. We hereafter omit the iteration index k for brevity. Ideally, the updated action μ^* is computed as the weighted average of candidate actions using weights $w(a) = \exp(f_{\hat{s}}(a))$. We observe that the SNIS ratio is invariant to a constant baseline shift. Dividing $w(a)$ by $\exp(f_{\hat{s}}(\mu))$, we define the baseline subtracted weight $\tilde{w}(a) := \exp(f_{\hat{s}}(a) - f_{\hat{s}}(\mu))$:

$$\mu^* = \sum_{i=1}^N \frac{\exp(f_{\hat{s}}(a_i)) \cdot a_i}{\sum_{j=1}^N \exp(f_{\hat{s}}(a_j))} = \sum_{i=1}^N \frac{\tilde{w}(a_i) \cdot a_i}{\sum_{j=1}^N \tilde{w}(a_j)}. \quad (130)$$

$\exp(f_{\hat{s}}(a) - f_{\hat{s}}(\mu))$ contains a log-probability difference term $\log \tilde{\pi}(a | s) - \log \tilde{\pi}(\mu | s)$, allowing us to directly apply the CRN technique described above.

806 D.4 Learning the Covariance Network

Covariance Σ_ψ in Eq. (8) is learnable via second moment matching, leveraging the target mean estimates from D.3. We assume diagonal covariance, where a neural network predicts $\sigma = f_\psi(s) \in \mathbb{R}^{|\mathcal{A}|}$ and equivalently $\Sigma = \sigma^2 I \in \mathbb{R}^{|\mathcal{A}| \times |\mathcal{A}|}$. The target variance of the d -th action as the weighted second moment is given by

$$(\sigma_d^*)^2 \approx \sum_{i=1}^N \frac{\bar{w}(a_i) (a_{i,d} - \mu_d^*)^2}{\sum_{j=1}^N \bar{w}(a_j)}. \quad (131)$$

Preventing Deterministic Collapse via Entropy Regularization. When the temperature α becomes small, the normalized weights $\bar{w}(a_i)$ can collapse toward a one-hot distribution, causing $\sigma_i \rightarrow 0$. This drives the covariance network toward a nearly deterministic policy and eliminates local exploration. To counteract this effect, we add an explicit entropy-preserving force based on the entropy of the local Gaussian policy. For $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_{|\mathcal{A}|}^2)$, the entropy of the local policy is $\mathcal{H}_{\text{local}} = \sum_{i=1}^{|\mathcal{A}|} \log \sigma_i + \text{const.}$, whose derivative w.r.t. σ_i is $1/\sigma_i$. Combining this entropy regularization term with the term induced by moment matching yields the following update:

$$\nabla_{\sigma_i} \mathcal{J}_{\text{total}} \approx \underbrace{\frac{(\sigma_i^*)^2 - \sigma_i^2}{\sigma_i^3}}_{\text{second moment matching}} + \underbrace{\beta \frac{1}{\sigma_i}}_{\text{entropy regularization}}. \quad (132)$$

The equilibrium yields $\sigma_i^2 = \frac{(\sigma_i^*)^2}{1-\beta}$ where larger $\beta \in [0, 1)$ enlarges the target variance relative to the pure moment matching and helps prevent variance collapse.

Log-Space Target Matching. In practice, the covariance network predicts the log-standard deviation $\varsigma_i = \log \sigma_i$. Applying the chain rule gives

$$\frac{\partial \mathcal{J}_{\text{total}}}{\partial \varsigma_i} = \frac{\partial \mathcal{J}_{\text{total}}}{\partial \sigma_i} \frac{\partial \sigma_i}{\partial \varsigma_i} = \frac{(\sigma_i^*)^2}{\sigma_i^2} - 1 + \beta. \quad (133)$$

Eq. (133) identifies the desired equilibrium, but directly optimizing this asymmetric gradient can be numerically unstable in bounded parameterizations of ς_i . Instead, we construct a surrogate loss whose minimizer matches the same equilibrium in log-space. From the equilibrium $\sigma_i^2 = \frac{(\sigma_i^*)^2}{1-\beta}$, the corresponding target log standard deviation is $\varsigma_{\text{target},i} = \frac{1}{2} \log((\sigma_i^*)^2) - \frac{1}{2} \log(1-\beta)$. To avoid the singularity at $(\sigma_i^*)^2 = 0$, we introduce a minimum variance floor $\sigma_{\min}^2$ and define the stabilized exploitation target covariance $\hat{\Sigma}_i = \max((\sigma_i^*)^2, \sigma_{\min}^2)$. We then compute the bounded target

$$\varsigma_{\text{target},i} = \text{clip}\left(\frac{1}{2} \log \hat{\Sigma}_i - \frac{1}{2} \log(1-\beta), \varsigma_{\min}, \varsigma_{\max}\right). \quad (134)$$

Finally, we update the covariance network using the symmetric log-space MSE objective

$$\mathcal{L}(\psi) = \mathbb{E}_{s \sim \mathcal{D}} \left[\frac{1}{D} \sum_{i=1}^D \left(\varsigma_i(s) - \text{sg}(\varsigma_{\text{target},i}(s)) \right)^2 \right], \quad (135)$$

where $\text{sg}(\cdot)$ is a stop-gradient operator.

Algorithm 1 FLAG: Flow Policy MaxEnt-RL by Latent Augmented Guidance

```
1: Initialize critic networks  $Q_{\phi_1}, Q_{\phi_2}$ , and vector field network  $u_\theta$  with random parameters  $\phi_1, \phi_2, \theta$ .
2: Initialize environment replay buffer  $\mathcal{D}_{\text{ENV}}$  and small guidance buffer  $\mathcal{D}_{\text{FLAG}}$ 
3: for  $k = 1$  to  $M$  do
4:   if  $k \% \text{UTD} = 0$  then
5:     Sample  $a \sim \pi(a \mid s; \theta_k)$ 
6:     Step environment:  $s' \sim p(s' \mid s, a)$ 
7:     Store  $(s, a, r, s')$  in  $\mathcal{D}_{\text{ENV}}$ 
8:   end if
9:   Sample  $B$  transitions  $(s, a, s', r) \sim \mathcal{D}$ 
10:  for each update step do
11:    Sample  $B$  transitions  $(s, a, r, s')$  from  $\mathcal{D}$ 
12:    Sample  $a' \sim \pi(a' \mid s'; \theta_k)$ 
13:    Compute the log probability of  $\tilde{\pi}$  (Eq. (28))
14:    Update critics:  $\phi \leftarrow \phi - \eta_\phi \nabla_\phi \mathcal{L}(\phi)$  (Eq. (128))
15:    if  $K \% \text{POLICY DELAY}$  then
16:      Sample  $N$  latent vectors  $z_1, \dots, z_N \sim p_z(z)$  and transport to actions  $\{T_{\theta_k}(s, z_i)\}_{i=1}^N$ 
17:      Compute the log probability of  $\tilde{\pi}$  (Eq. (28))
18:      Estimate  $f_{\hat{s}, k}(a_i)$  (Eq. (20)) and normalize with  $\alpha$ 
19:      Estimate  $\mu^*$  (Eq. (22)) and Update base flow policy:  $\theta_{k+1} \leftarrow \theta_k - \eta_k \nabla_\theta \mathcal{L}(\theta)$  (Eq. (25))
20:      Store  $(s, z, \mu^*)$  in  $\mathcal{D}_{\text{FLAG}}$ 
21:      Update  $\alpha_{k+1} \leftarrow \alpha_k - \eta_\alpha \mathcal{J}(\alpha)$  (Eq. (27))
22:    end if
23:    Sample  $B$  augmented-state-action pairs  $(s, z, \mu) \sim \mathcal{D}_{\text{FLAG}}$ 
24:    Update base flow policy:  $\theta_{k+1} \leftarrow \theta_k - \eta_\theta \nabla_\theta \mathcal{L}(\theta)$  (Eq. (25))
25:  end for
26: end for
```

E Baseline

E.1 Baselines in Section 5.1

MaxEntDP⁴ utilizes the Q-weighted Noise Estimation (QNE) method to approximate the exponential of the target distribution of MaxEnt-RL. In the original paper, They employ $N = 500$ action samples to ensure a low-variance and accurate training target for the noise prediction network, which is expensive. The policy naturally balances local and global update characteristics because the diffusion time step t controls the noise schedule α_t , which directly determines the scale of the sampled action area a_t during the training process. Our implementation follows their official Github repository.

DPMD⁵ utilizes **Rewighted Score Matching (RSM)** to optimize diffusion policies without requiring direct samples from the optimal policy. The algorithm weights the score matching objective with the exponential of the Q -function, $\exp(Q(s, a)/\lambda)$. To ensure efficient exploitation and manage the inherent randomness of the diffusion process, DPMD adopts batch action sampling—a form of soft rejection sampling—where the action $a = \arg \max_i Q(s, a_{(i)})$ is selected from numerous generated candidates ($P = 32$) for environment interaction, whereas $N = 1$ action is sampled in the policy update. The inference time is more expensive than FLAG, because we use "policy delay" for computational efficiency in policy update, while the batch action sampling cannot. To match the sampling budget with FLAG, we modify the original DPMD so that the algorithm does not leverage batch action sampling heuristics ($P = 1$) and use multiple action samples ($N \leq 64$) during policy update. Our implementation follows their official Github repository.

QVPO⁶ derives its policy update by leveraging a lower bound on the policy gradient objective. Specifically, rather than directly maximizing $\mathbb{E}_{a \sim \pi_\theta}[Q(s, a)]$ via gradient ascent, QVPO shows that this objective admits a tractable lower bound expressible as a Q-weighted score matching loss,

⁴<https://github.com/diffusionyes/MaxEntDP>

⁵https://github.com/mahaitongdae/diffusion_policy_online_rl

⁶<https://github.com/wadx2019/qvpo>

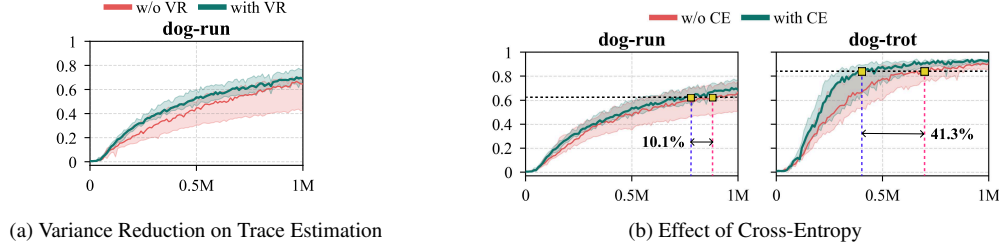


Figure 7: **Variance Reduction and Cross-Entropy Ablations.** (a) Training stability with and without the variance reduction technique for Hutchinson’s trace estimator. (b) We compare FLAG against a variant trained without the cross-entropy term. The horizontal dashed line represents 90% of peak performance, with intersection points marked by ■.

enabling the use of a diffusion model as the policy. Concretely, at each update step, $N = 64$ candidate actions are drawn from the current diffusion policy for each state, and the top- k ($k = 1$) actions by the minimum of two Q-networks are retained as positive targets. Only candidates whose Q-value exceeds a threshold ($Q > 1.0$) receive a nonzero weight, acting as a quality filter. The diffusion policy is then trained via the weighted denoising score matching objective. For exploration, QVPO augments the training batch with $N_{\text{neg}} = 10$ actions per positive sample drawn uniformly at random from $\mathcal{U}(-1, 1)$; these negative samples are assigned weights $w_i = \alpha \cdot Q_{\text{pos}}$ where α follows a linear decay schedule from 0.02 to 0.002 over 2×10^5 steps, encouraging broad exploration early in training. Our implementation follows their official Github repository.

For CrossQ update and distributional critic architecture we follows DIME official Github.

E.2 Baselines in Section 5.2

DIME [11]. For MuJoCo tasks, we run DIME⁷ using the implementation released in the official Github repository. For DMC Dog and MyoSuite tasks, we directly use the results reported in the original paper, together with the reported CrossQ and QSM baselines.

QSM [38]. For MuJoCo tasks, we run QSM⁸ using the official implementation. For a fair comparison, we align its hyperparameters with those used in DIME whenever applicable.

DACERv2 [48] and DIPO [49]. Our implementations of DACERv2⁹ and DIPO¹⁰ are based on their official Github repositories

Following DIME, we additionally incorporate the CrossQ update for the critic and use a distributional critic architecture.

FlowRL [26]. Our implementation of FlowRL¹¹ follows the official Github repository. FlowRL uses layer normalization [4] in the critic architecture and employs a three-layer MLP with hidden dimensions [512, 512, 512], which is architecturally different from the wider CrossQ critic with hidden dimensions [2048, 2048]. Therefore, we do not apply the CrossQ modification to FlowRL and instead retain the original critic architecture.

All hyperparameters are listed in Table 6.

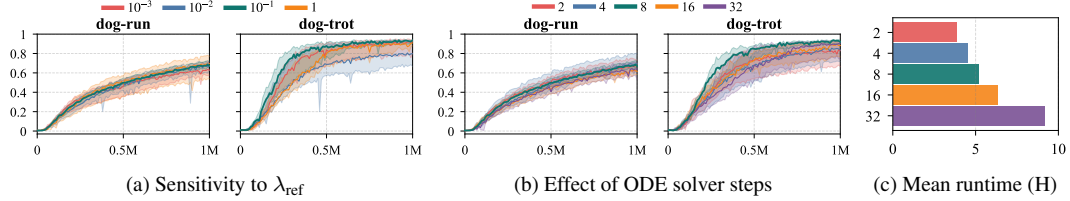


Figure 8: **Ablation and Efficiency Analysis.** (a) Performance curves across different values of the KL-divergence constraint parameter λ_{ref} . (b) Aggregate performance as a function of the number of integration steps used during flow inference. (c) Runtime comparison.

F Additional Ablation Studies

F.1 Variance Reduction in Hutchinson’s Trace Estimator

To efficiently estimate the log-likelihood of the flow policy, we employ Hutchinson’s trace estimator. To mitigate the stochasticity of this approximation, we adopt a variance reduction technique that shares the random tangent vectors across the action samples, as detailed in Appendix D.3. We ablate this design choice to quantify its impact on training stability (Figure 7a). The results demonstrate that this variance reduction strategy not only accelerates learning but also significantly reduces the variance across random seeds, leading to more robust convergence.

F.2 Cross-entropy.

Although MaxEnt RL promotes systematic exploration, exact entropy computation for expressive policies is infeasible. We circumvent this by introducing a cross-entropy term as a tractable surrogate. To verify this design, we conduct an ablation study comparing our approach with a standard formulation that lacks the MaxEnt component. We omit the log-probability from the soft Q-function update (Eq. 16) and the weight calculation (Eq. 21). The model with cross-entropy reaches the 90% of peak performance approximately 10.1% and 41.3% faster in respective tasks compared to the baseline in Figure 7b. This shows that the inclusion of cross-entropy regularizes the policy by preventing overfitting to early value overestimation, as evidenced by the reduced variance across random seeds.

F.3 Sensitivity to λ_{ref}

We examine the sensitivity of FLAG to the hyperparameter λ_{ref} , reporting performance on the DMC dog-run and dog-trot tasks in Figure 8a. This parameter controls the allowable divergence between the updated local policy and the reference policy; intuitively, a higher λ_{ref} permits more aggressive updates, while a lower value enforces a tighter trust region. As illustrated, performance is sensitive to this choice. Extreme values—either overly restrictive or excessively unstable—degrade performance. For our main experiments, we found $\lambda_{\text{ref}} = 10$ to yield the most consistent results across tasks.

F.4 Influence of ODE Solver Steps

We analyze the trade-off between inference precision and computational cost by varying the number of ODE solver steps used for the flow policy. The results are summarized in Figure 8b. This result shows that the performance is relatively robust to the number of ODE solver steps. Since the training duration increases proportionally with the number of steps (Figure 8c), we fixed the solver steps to 8 for all experiments to achieve computational efficiency.

⁷<https://github.com/ALRhub/DIME>

⁸https://github.com/escontra/score_matching_rl

⁹<https://github.com/happy-yan/DACER-Diffusion-with-Online-RL>

¹⁰<https://github.com/BellmanTimeHut/DIPO>

¹¹<https://github.com/bytedance/FlowRL>

Table 4: Environment state and action space dimensions

	Ant-v5	HalfCheetah-v5	Humanoid-v5	Walker2d-v5	DMC dog	MyoSuite
$\dim(\mathcal{S})$	105	17	348	17	223	93
$\dim(\mathcal{A})$	8	6	17	6	38	39

Table 5: Hyperparameters of experiments in Figure 3

	SDAC	DPMD	QVPO	MaxEntDP	FLAG
H_{target}	$-0.9\dim(\mathcal{A})$	$-0.9\dim(\mathcal{A})$	N/A	N/A	$-0.9\dim(\mathcal{A})$
Temp. Learn. Rate	$7e-3$	$7e-3$	N/A	N/A	$1e-3$
Critic Learn. Rate	$3e-4$	$3e-4$	$3e-4$	$3e-4$	$3e-4$
Actor/Score Learn. Rate	$3e-4 \rightarrow 3e-5$	$3e-4 \rightarrow 3e-5$	$3e-4 \rightarrow 3e-5$	$3e-4$	$3e-4$
Diffusion Steps	20	20	20	20	8
Discount	0.99	0.99	0.99	0.99	0.99
Batch size	256	256	256	256	256
Buffer size	$1e6$	$1e6$	$1e6$	$1e6$	$1e6$
Critic Hidden Depth	2	2	3	2	2
Critic Hidden Size	2048	2048	2048	2048	2048
Critic CrossQ	True	True	True	True	True
Critic LayerNorm	False	False	False	False	False
Dist. Critic Bins	101	101	101	101	101
Actor/Score Depth	3	3	3	3	3
Actor/Score Size	256	256	256	256	256
Update-to-data Ratio	2	2	2	2	2
Policy delay	3	3	3	3	3
Exploration Steps	10000	10000	10000	10000	10000

G Experiment Details

We conduct all experiments using JAX [9]. For FlowRL [26], we follow their implementation with PyTorch [35].

G.1 Multigoal Environment

We conduct a small-scale experiment in the MultiGoalEnv, adapted from [19], to qualitatively assess whether SDAC, DPMD, QVPO, MaxEntDP, and FLAG can recover a multimodal action distribution from a shared value function. An optimal Q-function $Q(s, a)$ is first computed via dynamic programming on a discrete grid and then frozen across all methods as a common scoring oracle, isolating policy extraction from representation learning. Each method optimizes a diffusion or flow matching policy to approximate the Boltzmann target $\pi^*(a | s) \propto \exp(Q(s, a)/\tau)$ with $\tau = 0.1$, and we visualize the learned action distributions at two probe states, $(0, 0)$ and $(3, 3)$, via

Table 6: Hyperparameters of experiments in Table 2

	DIME	FlowRL	DACERv2	QSM	DIPO	CrossQ	FLAG
H_{target}	$4\dim(\mathcal{A})$	N/A	$-0.9\dim(\mathcal{A})$	N/A	N/A	$-\dim(\mathcal{A})$	$-\dim(\mathcal{A})$
Temp. Learn. Rate	$1e-3$	N/A	$3e-2$	N/A	N/A	$3e-4$	$1e-3$
Critic Learn. Rate	$3e-4$	$3e-4$	$1e-4$	$3e-4$	$3e-4$	$7e-4$	$3e-4$
Actor/Score Learn. Rate	$3e-4$	$3e-4$	$1e-4$	$3e-4$	$3e-4$	$7e-4$	$3e-4$
Diffusion Steps	16	1	20	15	100	N/A	8
Discount	0.99	0.99	0.99	0.99	0.99	0.99	0.99
Batch size	256	256	256	256	256	256	256
Buffer size	$1e6$	$1e6$	$1e6$	$1e6$	$1e6$	$1e6$	$1e6$
Critic hidden depth	2	3	2	2	2	2	2
Critic hidden size	2048	512	2048	2048	2048	2048	2048
Critic use CrossQ	True	False	True	True	True	True	True
Critic use LayerNorm	False	True	False	False	False	False	False
Num. Bin/Quantiles	101	N/A	2	N/A	101	N/A	101
Actor/Score depth	3	2	3	3	4	3	3
Actor/Score size	256	512	256	256	256	256	256
Update-to-data ratio	2	1	1	1	2	2	2
Policy delay	3	1	1	1	3	3	3
Exploration Steps	5000	10000	10000	10000	10000	5000	10000
Prior Distr.	$\mathcal{N}(0, 2.5)$	$\mathcal{N}(0, 1)$	$\mathcal{N}(0, 1)$	$\mathcal{N}(0, 1)$	$\mathcal{N}(0, 1)$	N/A	$\mathcal{N}(0, 1)$
Optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam
Score-Q align. factor	N/A	N/A	N/A	50	N/A	N/A	N/A

Table 7: List of FLAG-specific hyperparameters

α_{init}	λ_{ref}	$\log \sigma_{\text{init}} / \log \sigma_{\text{min}}$	$\log \sigma$ Warmup / Decay Steps
0.01	10	-2 / -3	$2e5 / 8e5$
Supervision Warmup / Ramp Steps	Flow Buffer Size	Number of z given s in update	# of samples for Variance Reduction
$1e5 / 1e5$	10240	1	1

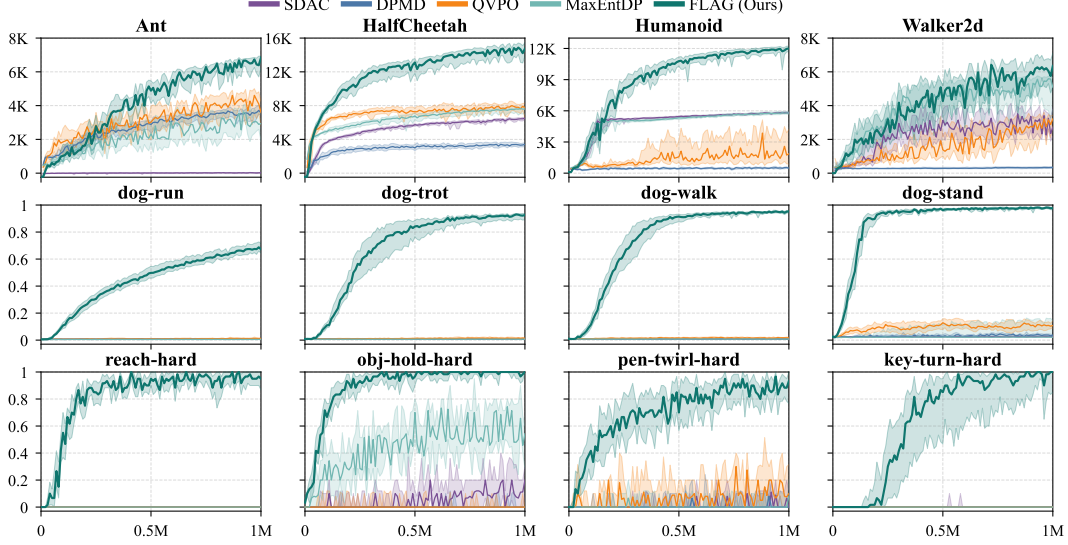


Figure 9: Full learning curves of Figure 3 (P=1)

kernel density estimation. By fixing the Q-function, this setup directly measures each method’s ability to capture the target distribution as a function of the number of importance samples $N \in \{2, 8, 32\}$ per gradient step, without confounding factors from Q-learning or environment interaction.

G.2 Comparison with Target Matching Methods

Figure 3. Scores are averaged over four tasks per benchmark: HalfCheetah, Ant, Walker2d, and Humanoid for MuJoCo; Dog-run, Dog-trot, Dog-walk, and Dog-stand for DMC Dog; and reach-hard, obj-hold-hard, key-turn-hard, and pen-twirl-hard for MyoSuite. Environment state and action space dimensions are summarized in Table 4. All methods are evaluated at 1M environment steps using 5 evaluation episodes, with IQM returns and confidence intervals computed over 10 seeds for $P = 1$ and 5 seeds for $P = 32$ due to the computational burden. GPU hours are measured on a single NVIDIA L40S GPU. Hyperparameters are reported in Table 5, and FLAG-specific hyperparameters shared across all experiments are listed in Table 7. Full learning curves corresponding to Figure 3 are provided in Figures 9 and 10.

Figure 4 We report performance at 1M steps using 5 evaluation episodes, with mean and standard deviation over 10 seeds for $P = 1$ and 5 seeds for $P = 32$, consistent with Figure 3. Hyperparameters follow those used for Figure 3 and Table 2, except for CrossQ usage and critic depth and hidden size. Specifically, we use $[256, 256, 256]$ critic architectures without Layer Normalization or Batch Renormalization. Please see Table 8 for detailed results.

G.3 Comparisons with Diffusion and Flow-based Algorithms

Table 2. We report performance at 1M environment interaction steps using 5 evaluation episodes with 10 random seeds per algorithm. Hyperparameters are reported in Table 6, and additional learning curves for the four MuJoCo tasks are provided in Figure 11.

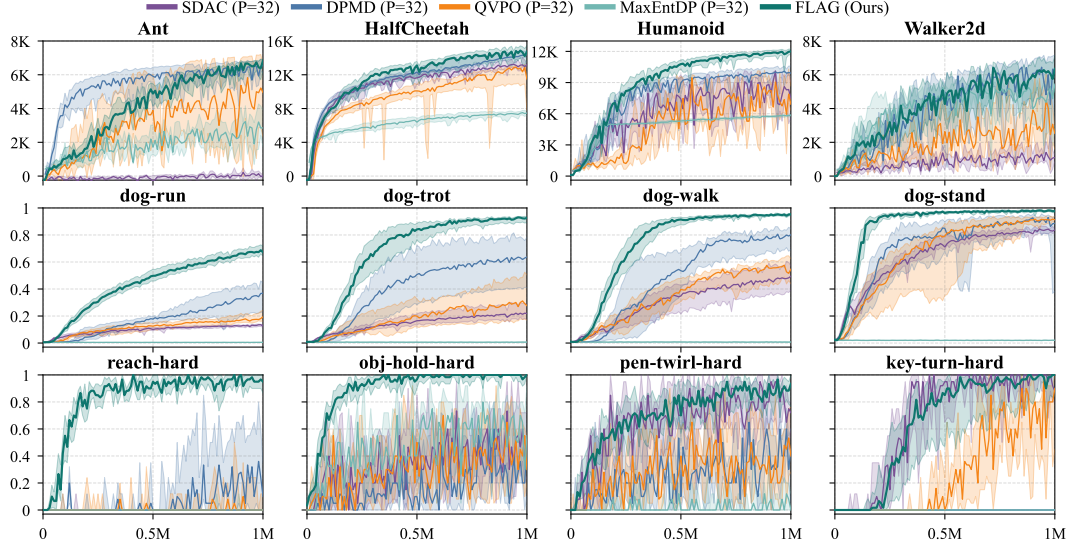


Figure 10: Full learning curves of Figure 3 (P=32)

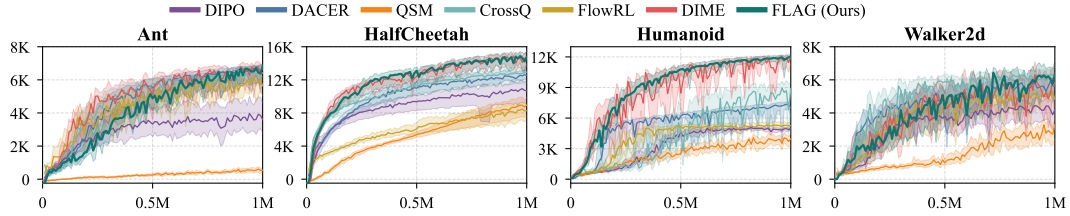


Figure 11: Learning curves other diffusion & flow-matching algorithms in 4 MuJoCo tasks

Table 8: DMC Dog performance without CrossQ in Section 5.1. Δ_F denotes the relative performance drop from FLAG, defined as $(1 - \text{Ret}_{\text{Alg}}/\text{Ret}_{\text{FLAG}}) \times 100$. $\dagger$ denotes the baselines’ policy update relevant default hyperparameters used in the original papers. The highest and second scores are highlighted.

P	N	Alg.	Dog-trot		Dog-run	
			Return (1k) $\uparrow$	$\Delta_F \downarrow$	Return (1k) $\uparrow$	$\Delta_F \downarrow$
1	64	SDAC	0.008 $\pm$ 0.001	0.984	0.006 $\pm$ 0.000	0.980
	64	DPMD	0.013 $\pm$ 0.008	0.974	0.006 $\pm$ 0.002	0.980
	64	QVPO	0.020 $\pm$ 0.022	0.960	0.039 $\pm$ 0.036	0.872
	500	MaxEntDP $\dagger$	0.007 $\pm$ 0.001	0.986	0.005 $\pm$ 0.001	0.984
32	64	SDAC $\dagger$	0.019 $\pm$ 0.023	0.962	0.007 $\pm$ 0.002	0.977
32	1	DPMD $\dagger$	0.117 $\pm$ 0.125	0.766	0.023 $\pm$ 0.016	0.924
4	64	QVPO $\dagger$	0.077 $\pm$ 0.082	0.846	0.045 $\pm$ 0.054	0.852
–	–	DIME $\dagger$	0.088 $\pm$ 0.073	0.824	0.025 $\pm$ 0.033	0.918
–	–	DACERv2 $\dagger$	0.107 $\pm$ 0.043	0.786	0.065 $\pm$ 0.059	0.786
1	8	FLAG $\dagger$ (ours)	0.500 $\pm$ 0.026	0.0	0.304 $\pm$ 0.027	0.0

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1145 Figure 8c provides a runtime bar chart comparing FLAG across different numbers of ODE

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1229 itories are linked in Appendix E.1. The benchmark environments DMC [46], MyoSuite [10],
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